POINTS OF PRIME ORDER ON ELLIPTIC CURVES OVER NUMBER FIELDS OF DEGREE AT MOST 22

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Abstract. Let $S(d)$ be the set of primes p for which there exists an elliptic curve over a number field of degree d with a point of order p. The sets $S(d)$ are known for $d \leq 8$. We determine $S(d)$ for all $9 \leq d \leq 22$. For example, the largest prime in $S(22)$ is 67, and $S(16)$ consists of the primes up to 37. To reduce the problem to finitely many primes we prove a version of Oesterlé's bound which uses reduction modulo 2 instead of modulo 3, and which is much smaller for $d \geq 13$. The remaining primes are dealt with using results of Derickx and Stoll and extensive computations with the modular curves $X_1(p)$ over $\mathbb{F}_2$ for $p \leq 163$.

1. Introduction

Let K be a number field and E/K an elliptic curve. By the Mordell-Weil theorem the torsion group $E(K)_{\text{tors}}$ is finite, and Merel [26] proved that its order is bounded by a constant depending only on the degree $d = [K : \mathbb{Q}]$. A key step in all the proofs of such results is to bound the primes that divide $\#E(K)_{\text{tors}}$. Following [17, 9] we denote by $S(d)$ the set of primes p for which there exists an elliptic curve over a number field of degree d with a point of order p, and by $\text{Primes}(x)$ the set of primes $p \leq x$.

Mazur [22] proved that $S(1) = \text{Primes}(7)$, Kamienny [16], building on work of Kenku and Momose [20], proved that $S(2) = \text{Primes}(13)$, and Parent [31, 32] proved that $S(3) = \text{Primes}(13)$. Derickx, Kamienny, Stein and Stoll [9, Theorem 1.2] proved that

$$S(4) = \text{Primes}(17), \quad S(5) = \text{Primes}(19), \quad S(6) = \text{Primes}(19) \cup \{37\}, \quad S(7) = \text{Primes}(23),$$

and Khawaja [21] proved that $S(8) = \text{Primes}(23)$. An important ingredient of [9] is a bound of Oesterlé, which states that $p \leq (3^{d/2} + 1)^2$ for all $p \in S(d)$ [9, Theorem 6.1]. Derickx and Stoll [11] studied the behavior of $S(d)$ as $d \to \infty$. Among other things they gave a criterion [11, Corollary 7.5], in terms of the *strange* characters modulo p, that rules out most primes below Oesterlé's bound, and they determined the strange characters modulo all primes $p < 10^5$ [11, Table 2]. In an unpublished preprint [12] they also determine $S(9)$. The sets $S(d)$ are the starting point of the classifications of the possible torsion groups of elliptic curves over number fields of degree d, which are now known for $d \leq 9$ (see [8, 10, 27, 28] and the references therein).

In this paper we determine $S(d)$ for $9 \leq d \leq 22$. Our main result is the following theorem.

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Theorem 1.1. *We have*

$$\begin{aligned} S(9) = S(11) &= \mathrm{Primes}(31), \\ S(10) = S(13) = S(16) &= \mathrm{Primes}(37), \\ S(12) = S(15) &= \mathrm{Primes}(37) \cup \{43\}, \\ S(14) = S(17) = S(18) = S(19) &= \mathrm{Primes}(43), \\ S(20) &= \mathrm{Primes}(47) \cup \{61\}, \\ S(21) &= \mathrm{Primes}(47), \\ S(22) &= \mathrm{Primes}(53) \cup \{67\}. \end{aligned}$$

Together with the results mentioned above, this determines $S(d)$ for all $d \leq 22$. Note that $S(d') \subseteq S(d)$ if d' divides d, but that $S(d)$ is not increasing in d: for example $37 \in S(10)$, but $37 \notin S(11)$, and $41, 43 \in S(14)$, but $41, 43 \notin S(16)$. The primes $31 \in S(10)$, $29, 43 \in S(14)$, $41, 61 \in S(20)$ and $67 \in S(22)$ are of the form $2d + 1$ or $3d + 1$, and such primes are always in $S(d)$ for even d, because of points with complex multiplication [11, Corollary 7.5]. Derickx and Stoll [11, Corollary 7.8] showed that, assuming a conjecture on the sparsity of strange characters, $S(d)$ is contained in $\mathrm{Primes}(d + 1) \cup \{\frac{3}{2}d + 1, 2d + 1, 3d + 1\}$ for d large enough. Note that the degrees we consider are far from this range, as $S(22)$ contains all the primes up to 53.

We now describe the proof. For a point of order p on an elliptic curve over a number field of degree d, we study the corresponding effective divisor D of degree d on the modular curve $X_1(p)$, and our methods depend on the size of p.

For large p we use Oesterlé's bound together with [11, Corollary 7.5]. Unfortunately, Oesterlé's bound exceeds 10^5, the limit of [11, Table 2], for $d \geq 11$, and it grows like 3^d. For $d \geq 13$ we prove a version of Oesterlé's bound that uses reduction modulo 2 instead of modulo 3 (Proposition 4.4), which gives bounds of size about 2^d. The price we pay for using the prime 2 is that we have to check that $2d$, instead of d, Hecke operators acting on the winding element are linearly independent modulo 2. For $d \in \{11, 12\}$ we use reduction modulo 3, 5 and 7 instead. Above 10^5 at most 1488 primes remain in each degree, and for each of them we show that sufficiently many characters are not strange, by a computation with modular symbols over a finite field (Section 4.3).

For medium-sized p we use [11, Proposition 4.1], which shows that D is invariant under a diamond operator whenever the gonality of $X_1(p)$ is large compared to d (Section 4.4). For a few primes $p > 157$ its hypothesis on rational 2-torsion is not known, and we check it by showing that a Hecke operator restricts to an isogeny of odd degree on a suitable abelian subvariety of $J_1(p)$.

The remaining primes, which satisfy $37 \leq p \leq 163$, require most of the work (Sections 5 and 6). We reduce D modulo 2 and enumerate all the possible reductions, using all the elliptic curves over $\mathbb{F}_{2^f}$ for $f \leq d$. Reductions supported on rational cusps are ruled out by the formal immersion criterion of [9]. For all other reductions we use the Hecke operator $A_3 = T_3 - 1 - 3\langle 3 \rangle$, composed with an element of the group ring of the diamond operators which kills the part of $J_1(p)$ of positive rank: if D exists, a certain divisor of degree 0 on $X_1(p)_{\mathbb{F}_2}$ has to be principal [9, Lemma 8.6]. We show that it is not, by linear algebra in the canonical ring of $X_1(p)_{\mathbb{F}_2}$, which we compute from q-expansions. This argument rules out all but a short list of reductions. For these we show that D is invariant under a diamond operator, which typically leads to a contradiction by counting, or we use a relative version of the formal immersion criterion, or the formal immersion criterion at the reductions of known points of smaller degree, or quotients of $X_1(p)$.

Our methods are those of [9], [11] and [21], applied to a much larger range of degrees. The main difference is the scale of the computations: for $d = 22$ we have to deal with 155750 primes, the local computations concern 514 types of reductions at 18 primes, and they consist of 134762 new linear algebra certificates on the curves $X_1(p)_{\mathbb{F}_2}$, some of genus above 1000. The certificates for $d = 22$ alone occupy about 220 GB.

All the code used in the paper can be found at

https://github.com/F-Najman/Sd

AI disclosure and comments. AI models were extensively used in the work leading to this paper. OpenAI's model produced almost all of the mathematical content of the first versions of the proofs and carried out the computations. Anthropic's Claude Opus 5.5 reviewed these proofs, without rerunning the computations, wrote the draft of the present manuscript and organized the code repository.

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2. Notation and auxiliary results

Throughout the paper $p \geq 5$ is a prime. The modular curve $X_1(p)$ parametrizes pairs (E, P), where E is an elliptic curve and $P \in E$ is a point of order p, and it is a fine moduli space. It has a smooth proper model over $\mathbb{Z}[1/p]$, which we also denote by $X_1(p)$, so in particular it has good reduction at 2. We write $J_1(p)$ for its Jacobian, $X_0(p)$ and $J_0(p)$ for the curve parametrizing pairs (E, C) with $C \subseteq E$ cyclic of order p and for its Jacobian, $X^{(d)}$ for the d-th symmetric power of a curve X, and $\mathrm{gon}_k X$ for the gonality of X over a field k. For an effective divisor D of degree d on a curve X over $\mathbb{Q}$ we write D also for the corresponding point of $X^{(d)}(\mathbb{Q})$, and, when X has good reduction at 2, $\overline{D}$ for its reduction modulo 2, an effective divisor of degree d on $X_{\mathbb{F}_2}$.

2.1. **Diamond and Hecke operators.** For $a \in (\mathbb{Z}/p\mathbb{Z})^\times$ we write $\langle a \rangle$ for the automorphism $(E, P) \mapsto (E, aP)$ of $X_1(p)$. It depends only on the class of a in

$$\Delta(p) := (\mathbb{Z}/p\mathbb{Z})^\times/\{\pm 1\},$$

which is a cyclic group of order $(p-1)/2$, and for $h \in \Delta(p)$ we denote by $\mathrm{ord}(h)$ its order in $\Delta(p)$. We identify $h \in \Delta(p)$ with the corresponding diamond operator, and for a subgroup $H \subseteq \Delta(p)$ we write $X_H = X_1(p)/H$. For a prime $q \neq p$ we denote by T_q the Hecke correspondence

$$T_q(E, P) = \sum_C (E/C, P \bmod C),$$

the sum running over the $q + 1$ subgroups $C \subseteq E$ of order q, and we put

$$A_q = T_q - 1 - q\langle q \rangle.$$

Diamond and Hecke operators act on divisors on $X_1(p)$, on $J_1(p)$, and, by reduction, on divisors on $X_1(p)_{\mathbb{F}_2}$ when q is odd.

The curve $X_1(p)$ has $(p-1)/2$ cusps defined over $\mathbb{Q}$, which we call the *rational cusps*, and $(p-1)/2$ cusps defined over $\mathbb{Q}(\zeta_p)^+$, the maximal real subfield of $\mathbb{Q}(\zeta_p)$. The group $\Delta(p)$ acts simply transitively on each of these two sets. The rational cusps reduce to the $\mathbb{F}_2$-rational cusps of $X_1(p)_{\mathbb{F}_2}$, and the other cusps reduce to closed points of degree $\mathrm{ord}(2)$, which we call the *non-rational cusps*.

2.2. **The trace divisor.** Let K be a number field of degree d and E/K an elliptic curve with a point $P \in E(K)$ of order p. Then $x = (E, P)$ is a non-cuspidal point of $X_1(p)$ defined over K, and we call

$$D = \sum_{\sigma : K \hookrightarrow \overline{\mathbb{Q}}} \sigma(x)$$

the *trace divisor* of (E, P). It is an effective divisor of degree d defined over $\mathbb{Q}$, its support contains no cusps, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts transitively on its support, and all the points in its support have the same multiplicity d/n, where $n = [\mathbb{Q}(x) : \mathbb{Q}]$ divides d. In particular $p \in S(d)$ if and only if such a divisor exists. If $p \notin S(d')$ for every proper divisor d' of d, then $n = d$ and D is a closed point of degree d.

Lemma 2.1. *Let $x = (E, P)$ be a non-cuspidal point of $X_1(p)$ of degree n which is fixed by a non-trivial element of $\Delta(p)$. Then n is even, and either $j(E) = 1728$ and $n \geq (p-1)/2$, or $j(E) = 0$ and $n \geq (p-1)/3$.*

Proof. As $X_1(p)$ is a fine moduli space, we can take E and P to be defined over $F = \mathbb{Q}(x)$. If $\langle a \rangle \neq 1$ fixes x, there is an automorphism $\alpha \neq \pm 1$ of E with $\alpha(P) = aP$. It is unique, as an automorphism $\beta \neq 1$ fixes no point of order p ($\beta - 1$ has degree at most 4), so α is defined over F. Hence $j(E) \in \{0, 1728\}$, and α acts on the invariant differential by a root of unity $\zeta \in F$ of order 3, 4 or 6, so the field $k = \mathbb{Q}(\zeta)$ of complex multiplication of E is contained in F, and n is even. By a theorem of Silverberg [35] (see [33, Section 1] for the statement we use), as $k \subseteq F$ and $E(F)$ has a point of order p, we have $p - 1 \leq \frac{1}{2}\mu n$, where $\mu = \#\operatorname{Aut}(E)$ is 4 if $j(E) = 1728$ and 6 if $j(E) = 0$. □

Lemma 2.2. *Let D be the trace divisor of (E, P), let $n = [\mathbb{Q}(x) : \mathbb{Q}]$ be as above, and let $h \in \Delta(p)$ be such that $hD = D$. Then the orbits of the group generated by h on the (geometric) support of D all have the same length m, and m divides n. If n is odd or $n < (p-1)/3$, then $m = \mathrm{ord}(h)$.*

Proof. As h is defined over $\mathbb{Q}$, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ permutes the orbits of $\langle h \rangle$, and as it acts transitively on the n points of the support of D, all the orbits have the same length m, and $m \mid n$. The last statement follows from Lemma 2.1. □

By the proof of Lemma 2.1, the stabilizer in $\Delta(p)$ of a non-cuspidal point has order dividing 2 or 3, so in general $m \in \{\mathrm{ord}(h), \mathrm{ord}(h)/2, \mathrm{ord}(h)/3\}$. Note that for $p > 67$ and $n \leq 22$ we always have $n < (p-1)/3$.

2.3. **Finiteness of Mordell-Weil groups.** Recall from [11, Definition 5.1] that a non-trivial even Dirichlet character χ modulo p is *strange* if there is a newform f of weight 2 and level p with character χ such that $L(f, 1) = 0$, that p is *strange* if there is a strange character modulo p, and that $\mathrm{str}(p)$ is the order of the group generated by the strange characters modulo p. The strange characters modulo $p < 10^5$ are listed in [11, Proposition 5.2 and Table 2]. The strange primes below 200 are 61, 97, 101, 181 and 193, and their strange characters have orders 6, 12, 10, 6 and 12, respectively.

Lemma 2.3. *Let $\alpha \in \mathbb{Z}[\Delta(p)]$ be such that $\chi(\alpha) = 0$ for the trivial character χ and for all the strange characters χ modulo p. Then $\alpha J_1(p)(\mathbb{Q})$ is finite. In particular, if p is not strange, then $(h-1)J_1(p)(\mathbb{Q})$ is finite for every $h \in \Delta(p)$. If $p \in \{41, 47, 59, 71\}$, then $J_1(p)(\mathbb{Q})$ is finite.*

Proof. Up to isogeny over $\mathbb{Q}$, $J_1(p)$ is the product of the abelian varieties A_f attached to the Galois orbits of newforms f of weight 2 and level p, and $\Delta(p)$ acts on A_f through the character of f. Hence $\alpha J_1(p)$ is contained in the sum of the A_f for which the character of f is non-trivial

and not strange. These have $L(f,1) \neq 0$, so $A_f(\mathbb{Q})$ is finite by Kato's theorem [18, Corollary 14.3]. The last statement is [9, Proposition 3.1]. □

Lemma 2.4. a) *Let $A/\mathbb{Q}$ be an abelian variety with good reduction at a prime ℓ. If ℓ is odd, then reduction modulo ℓ is injective on $A(\mathbb{Q})_{\mathrm{tors}}$. If $\ell = 2$, then the kernel of reduction modulo 2 on $A(\mathbb{Q})_{\mathrm{tors}}$ is killed by 2.*

b) *For an odd prime $q \neq p$, the operator A_q kills $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$, and $A_q c = 0$ as a divisor for every rational cusp c. If $p \leq 157$, then A_2 kills $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$ as well.*

Proof. Part a) is [19, Appendix]. On $X_1(p)_{\mathbb{F}_q}$ we have $T_q = \mathrm{Frob}_q + q\langle q\rangle \mathrm{Frob}_q^{-1}$ (the Eichler-Shimura relation), so A_q kills $J_1(p)(\mathbb{F}_q)$, and as $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$ injects into $J_1(p)(\mathbb{F}_q)$ for odd q by a), A_q kills $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$. The action of T_q on the cusps is computed with Tate curves, see [11, Proposition 2.1]. Note that the modular interpretation used in [11] differs from ours by the Atkin-Lehner involution, which transforms $T_q - \langle q\rangle - q$ into A_q (see the discussion after [9, Lemma 8.6]). For $q = 2$ the same argument shows that A_2 kills the odd part of $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$, and for $p \leq 157$ the 2-primary part of $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$ is generated by differences of rational cusps [11, Remark 4.4], so it is killed by A_2 as well. □

2.4. **Gonality.** If a curve X over $\mathbb{Q}$ has good reduction at a prime ℓ, then $\mathrm{gon}_{\mathbb{Q}} X \geq \mathrm{gon}_{\mathbb{F}_\ell} X$, and an obvious count of points gives $\mathrm{gon}_{\mathbb{F}_\ell} X \geq \#X(\mathbb{F}_{\ell^k})/(\ell^k + 1)$ for all $k \geq 1$. We will use the following lower bounds.

Proposition 2.5. a) *We have $\mathrm{gon}_{\mathbb{Q}} X_1(p) \geq \frac{325}{2^{16}}(p^2 - 1)$, and if $p \leq 880$, then $\mathrm{gon}_{\mathbb{Q}} X_1(p) \geq \frac{1}{192}(p^2 - 1)$.*

b) *We have $\mathrm{gon}_{\mathbb{Q}} X_1(37) = 18$, $\mathrm{gon}_{\mathbb{Q}} X_1(43) = 24$ and $\mathrm{gon}_{\mathbb{Q}} X_1(53) \geq 20$.*

Proof. The first statement of a) is [11, Theorem 3.3]. The second follows in the same way from [2, Theorem 0.1] and the Selberg eigenvalue conjecture for $\Gamma_1(N)$ with $N \leq 880$, proved by Booker, Lee and Strömbergsson [5, Theorem 1.1]. The first two statements of b) are [13, Table 1] and [39, Theorem D].

The curve $X_1(53)_{\mathbb{F}_2}$ has 26 rational points, the reductions of the rational cusps, and no closed points of degree between 2 and 8. Let f be a function of degree $e \leq 19$ on $X_1(53)_{\mathbb{F}_2}$. One of the fibres of f above the three points of $\mathbb{P}^1(\mathbb{F}_2)$ contains at least 9 rational points, and after composing f with an automorphism of $\mathbb{P}^1$ we can assume that this is the fibre above ∞. Its non-rational part then has degree at most 10, so it is 0, a closed point of degree 9, or a closed point of degree 10. Hence the divisor of poles of f is bounded by $C + E$, where C is the sum of all the rational points and E is either an effective divisor of degree 10 supported on the rational points, or the sum of a rational point and a closed point of degree 9, or a closed point of degree 10. Using the action of $\Delta(53)$ we compute that $\ell(C + E) = 1$ in the first case, and also in the other two, so f does not exist. Hence $\mathrm{gon}_{\mathbb{Q}} X_1(53) \geq \mathrm{gon}_{\mathbb{F}_2} X_1(53) \geq 20$. □

2.5. **Two simple lemmas.** The following is trivial.

Lemma 2.6. *Let X be a curve over a field F, and let D_1 and D_2 be effective divisors of degree d on X, defined over F, such that $D_1 \sim D_2$. If $\mathrm{gon}_F X > d$, then $D_1 = D_2$.*

We will often use the following lemma, which is a variant of the argument of [11, Proposition 4.1].

Lemma 2.7. *Let D be the trace divisor of a point of order p over a number field of degree d, and write $\overline{D} = Z + C'$, where C' is supported on the $\mathbb{F}_2$-rational cusps and Z has no such points in its support. Let C be the rational cuspidal divisor lifting C', and let $c = \deg C$. Let $h \in \Delta(p)$ be such that $(h-1)J_1(p)(\mathbb{Q})$ is finite and $hZ = Z$, and put $y = [h(D-C)-(D-C)] \in J_1(p)(\mathbb{Q})$.*

a) *Then $2y = 0$, and $y = 0$ if one of the following holds:*
   i) *reduction modulo 2 is injective on $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$, for example if $J_1(p)(\mathbb{Q})[2] = 0$,*
   ii) *$\mathrm{ord}(h)$ is odd and h acts trivially on $J_1(p)(\mathbb{Q})[2]$,*
   iii) *$\mathrm{ord}(h)$ is odd and there is an endomorphism β of $J_1(p)$ which kills $J_1(p)(\mathbb{Q})_{\mathrm{tors}}$ and restricts to an isogeny of odd degree on the abelian subvariety $(h-1)J_1(p)$.*

b) *If $y = 0$ and $\mathrm{gon}_{\mathbb{Q}} X_1(p) > d + c$, or if $\mathrm{gon}_{\mathbb{Q}} X_1(p) > 2(d + c)$, then $hD = D$ and $hC = C$. In particular $\mathrm{ord}(h)$ divides c.*

Proof. Let $m = \mathrm{ord}(h)$, put $Z_0 = D - C$ and $N = 1 + h + \cdots + h^{m-1}$. Note that Z_0 need not have degree 0, but $mZ_0 - NZ_0$ does, and

$$my = (h-1)[mZ_0 - NZ_0],$$

so y is torsion. The reduction of y is the class of $hZ - Z = 0$, so $2y = 0$ by Lemma 2.4 a), and in case i) $y = 0$. In case ii) we have $Ny = my$, and $Ny = [NhZ_0 - NZ_0] = 0$, so $my = 0$, and as m is odd, $y = 0$. In case iii), as m is odd and $2y = 0$, we have $y = my = (m - N)y$, and as $m - N$ is divisible by $h - 1$ in $\mathbb{Z}[h]$, y lies in $B = (h-1)J_1(p)$. Since $\beta(y) = 0$ and β has odd degree on B, it follows that $y = 0$.

If $y = 0$, then $hD + C \sim D + hC$, and if $2y = 0$, then $2hD + 2C \sim 2D + 2hC$. These divisors are defined over $\mathbb{Q}$, so under the assumptions of b) Lemma 2.6 shows that $hD + C = D + hC$. As D has no cusps in its support, $hD = D$ and $hC = C$. Finally, $\Delta(p)$ acts freely on the rational cusps, so C is a sum of orbits of length m. □

2.6. **Non-principal divisors.** The following criterion, which we also used in [28], is our main tool for showing that a divisor of degree 0 over a finite field is not principal when the genus is large. For a curve X we write $R_i = H^0(X, iK_X)$, where K_X is a canonical divisor.

Lemma 2.8. *Let X be a smooth projective geometrically integral curve over a field k, and let $B = U - V$, where U, V are effective divisors of the same degree. Let $\mathcal{L}$ and $\mathcal{M}$ be line bundles on X, and choose $0 \neq F \in H^0(X, \mathcal{L}(-U))$ and $H \in H^0(X, \mathcal{M}(-U))$. If the linear map*

$$H^0(X, \mathcal{L}) \longrightarrow H^0(V, \mathcal{L}|_V) \oplus \frac{H^0(X, \mathcal{L} \otimes \mathcal{M})}{F \cdot H^0(X, \mathcal{M})}, \qquad G \longmapsto (G|_V,\ GH \bmod F \cdot H^0(X, \mathcal{M}))$$

is injective, then B is not principal. The same holds if several sections H are used.

Proof. Suppose $B = \mathrm{div}(g)$. Then $G = F/g$ is a non-zero section of $\mathcal{L}$ vanishing on V, and $GH = F \cdot (H/g)$ with $H/g \in H^0(X, \mathcal{M})$, so G is in the kernel of the map. □

We apply the lemma with $\mathcal{L} = \mathcal{M} = K_X$ or $2K_X$, where the multiplication maps $R_i \times R_j \to R_{i+j}$ are computed from the q-expansions of a basis of the weight 2 cusp forms, with enough coefficients to determine all the products. To evaluate sections at the points of V we use local parameters at these points, which we obtain by deforming the corresponding elliptic curve and lifting its point of order p by Hensel's lemma, together with the modular units of Sutherland [36] (see [13, Section 2]), whose linear relations with the canonical forms we verify to a precision beyond their degree bounds.

2.7. **Formal immersions.** We will use the formal immersion criterion in the following two forms. Let X be a smooth curve over $\mathbb{Z}_{(2)}$ (for us, $X_1(p)$ or X_H), J its Jacobian, t an endomorphism of J, and for $d \geq 1$ let

$$f : X^{(d)} \to J, \qquad E \mapsto t[E - dc_0],$$

where c_0 is a rational cusp. Let $\overline{x} \in X^{(d)}(\mathbb{F}_2)$. The cotangent map of f at $\overline{x}$ is the restriction of $t^*H^0(J_{\mathbb{F}_2}, \Omega^1) = t^*H^0(X_{\mathbb{F}_2}, \Omega^1)$ to the divisor $\overline{x}$, and f is a *formal immersion* at $\overline{x}$ if this cotangent map is surjective, i.e. if the restriction map

$$t^*H^0(X_{\mathbb{F}_2}, \Omega^1) \to H^0(X_{\mathbb{F}_2}, \Omega^1 \otimes \mathcal{O}_{\overline{x}})$$

is surjective. At a sum $\overline{x} = \sum_k d_k c_k$ of rational cusps, with $c_k = h_k c_0$ for distinct $h_k \in \Delta(p)$, this means that the vectors of the first d_k coefficients of the q-expansions of the forms in $t^*H^0(X_{\mathbb{F}_2}, \Omega^1)$ at the cusps c_k span $\mathbb{F}_2^d$, which is equivalent to the linear independence of the Hecke operators $t\, h_k T_j$, $1 \le j \le d_k$, modulo 2 [9, Proposition 5.3].

Lemma 2.9 ([9, Corollaries 4.3 and 4.4, Lemma 8.2])**.** *Suppose that $t(J(\mathbb{Q}))$ is finite and that reduction modulo 2 is injective on it (for example, $t(J(\mathbb{Q})) = 0$), and that f is a formal immersion at $\overline{x} \in X^{(d)}(\mathbb{F}_2)$. If $D_1, D_2 \in X^{(d)}(\mathbb{Q})$ both reduce to $\overline{x}$, then $D_1 = D_2$.*

Typically $t = \alpha A_q$, where $\alpha \in \mathbb{Z}[\Delta(p)]$ has $\alpha J_1(p)(\mathbb{Q})$ finite (Lemma 2.3) and q is an odd prime. Then $t(J_1(p)(\mathbb{Q})) = 0$ by Lemma 2.4. If $\overline{x}$ is a sum of rational cusps, D_2 is its rational lift, and Lemma 2.9 shows that no trace divisor reduces to $\overline{x}$. If $\overline{x}$ is the reduction of a known effective divisor D_2 which is not a trace divisor (for example the sum of a point of smaller degree and a rational cusp), it again shows that no trace divisor reduces to $\overline{x}$. Note that if f is a formal immersion at $\overline{x}$, then it is a formal immersion at every effective $\overline{x}' \le \overline{x}$, so a computation in degree d covers all the cuspidal reductions in smaller degrees: pad a divisor of degree $d' < d$ by adding $d - d'$ to one of its multiplicities.

When f is constant on a family of divisors through $\overline{x}$, for example on a family of pullbacks from a quotient of X, it cannot be a formal immersion at $\overline{x}$, and we use the following relative version, which is an integral form of the relative symmetric Chabauty method of Siksek [34].

Lemma 2.10. *Let S be a smooth scheme of relative dimension d over $\mathbb{Z}_2$, let $\overline{x} \in S(\mathbb{F}_2)$, and let $L \subseteq \widehat{S}_{\overline{x}}$ be a smooth closed formal subscheme of relative dimension r of the formal completion of S at $\overline{x}$. Let $\Phi : S \to A$ be a morphism to a smooth group scheme over $\mathbb{Z}_2$ with $\Phi(L) = 0$, and suppose that the kernel of the tangent map of $\Phi_{\mathbb{F}_2}$ at $\overline{x}$ is the tangent space of $L_{\mathbb{F}_2}$. Then every $D \in S(\mathbb{Z}_2)$ reducing to $\overline{x}$ with $\Phi(D) = 0$ lies on L.*

Proof. Choose formal coordinates $u_1, \dots, u_r, v_1, \dots, v_{d-r}$ at $\overline{x}$ such that L is given by $v = 0$, and functions $g_1, \dots, g_{d-r}$ on A vanishing at 0 such that the differentials of the $\Phi^* g_i$ at $\overline{x}$ are linearly independent, which is possible by the assumption on the tangent map. As the $\Phi^* g_i$ vanish on L, we can write $\Phi^* g_i = \sum_j m_{ij} v_j$, and the matrix (m_{ij}) is invertible at $\overline{x}$, hence invertible over the completed local ring. So the common zero locus of the $\Phi^* g_i$ is L. □

We apply Lemma 2.10 to $S = X^{(d)}$ and $\Phi = f$ with t as above, and with L of the form $L = R + \pi^* Y^{(r)}$, where $\pi : X \to Y = X/H$ is the quotient by a subgroup $H \subseteq \Delta(p)$ and R is a fixed effective divisor, typically a non-zero rational cuspidal divisor. Here L is smooth when H acts freely on the support of $\overline{x} - \overline{R}$, or more generally when the quotient map is tamely ramified there. If t also kills $\pi^* J_Y$ (for example if $t = (h-1)A_q$ with $h \in H$), then f vanishes on L. If the rank condition holds, a trace divisor reducing to $\overline{x}$ lies on L, so it contains R, which is impossible if R contains a cusp. In some cases R also contains the unique lifts of the points of $\overline{x}$ that are fixed by a subgroup of order 3 of $\Delta(p)$.

3. Points of prime order over number fields of degree $9 \le d \le 22$

In this section we show that the primes listed in Theorem 1.1 belong to $S(d)$.

Lemma 3.1. *If d' divides d, then $S(d') \subseteq S(d)$.*

Proof. Let K' be a number field of degree d' and E/K' an elliptic curve with a point of order p. Let ℓ be a prime which is unramified in K', and α a root of $x^{d/d'} - \ell$. This polynomial is Eisenstein at every prime of K' above ℓ, so $K = K'(\alpha)$ has degree d, and $E(K)$ has a point of order p. □

Lemma 3.2. *If $d \geq 2g(X_1(p))$, then $p \in S(d)$. In particular $11, 13, 17 \in S(d)$ for $d \geq 10$, and $19 \in S(d)$ for $d \geq 14$.*

Proof. Let c be a rational cusp of $X_1(p)$. As $d - 1 \geq 2g - 1$, the Riemann-Roch theorem gives $\ell(dc) = \ell((d-1)c) + 1$, so there is a function f on $X_1(p)$ whose only pole is a pole of order d at c. By Hilbert's irreducibility theorem, the fibre $f^{-1}(t)$ is a closed point of degree d for infinitely many $t \in \mathbb{Q}$, and all but finitely many of these points are not cusps. As $X_1(p)$ is a fine moduli space, each of them gives an elliptic curve with a point of order p over a number field of degree d. The second statement follows as $X_1(11)$, $X_1(13)$, $X_1(17)$ and $X_1(19)$ have genus 1, 2, 5 and 7. □

Most of our explicit examples are of the following form. Let $f \in \mathbb{Q}[t]$ be an irreducible polynomial of degree d, let x be a root of f, put $K = \mathbb{Q}(x)$, and let $y \in K$ be such that (x, y) is a point on Sutherland's model of $X_1(p)$ [36]. Put

$$r = \frac{x^2y - xy + y - 1}{x^2y - x}, \quad s = \frac{xy - y + 1}{xy}, \quad b = rs(r-1), \quad c = s(r-1).$$

Then $P = (0, 0)$ is a point of order p on the elliptic curve

$$E(b, c): \ Y^2 + (1 - c)XY - bY = X^3 - bX^2$$

over K. For each example we check directly that f is irreducible, that $E(b, c)$ is non-singular and that $pP = 0$, so we do not need to know where the examples came from. Most of these examples come from the tables of low degree points on $X_1(p)$ of van Hoeij [38], or from specializations of functions of degree d on $X_1(p)$. For example, for $p = 43$ and $d = 12$ we can take

$$\begin{aligned} f &= t^{12} - t^{11} - 5t^{10} + 6t^9 + 9t^8 - 16t^7 - 3t^6 + 19t^5 - 3t^4 - 12t^3 + t^2 + 4t + 1, \\ 43y &= 35x^{11} - 65x^{10} - 107x^9 + 314x^8 - 34x^7 - 580x^6 + 601x^5 \\ &\quad + 156x^4 - 552x^3 + 219x^2 + 136x - 81, \end{aligned}$$

and for $p = 37$ and $d = 10$ we can take

$$\begin{aligned} f &= t^{10} - 5t^8 + 3t^7 + 10t^6 - 8t^5 - 7t^4 + 9t^3 - 3t + 1, \\ y &= 2x^9 + x^8 - 10x^7 + x^6 + 23x^5 - 6x^4 - 22x^3 + 11x^2 + 9x - 5. \end{aligned}$$

Here $j(E)$ has degree 4 in the first example and degree 10 in the second. The few examples that are not of this form are similar. For instance, for $(d, p) = (22, 67)$ we use a point of order 67 on a twist of $y^2 = x^3 + 1$ over a field of degree 22, which comes from a point with complex multiplication.

Proposition 3.3. *For $9 \leq d \leq 22$, all the primes listed in Theorem 1.1 belong to $S(d)$.*

Proof. By Lemma 3.1, the results for $d \leq 8$ listed in the introduction, [29] for $p \in \{17, 19, 23\}$ and $d = 9$, and Lemma 3.2, it is enough to give an example over a number field of degree d for each of the pairs (d, p) in Table 1. Here we also use Lemma 3.1 for $d \in \{18, 20, 22\}$ with $d' = 9, 10, 11$, respectively. For each of these pairs we give an explicit example as above, and we verify that it has the required properties. □

d	p	d	p
9	$29, 31$	16	$29, 31, 37$
10	$23, 29, 31, 37$	17	$23, 29, 31, 37, 41, 43$
11	$19, 23, 29, 31$	18	$41, 43$
12	$23, 29, 31, 43$	19	$23, 29, 31, 37, 41, 43$
13	$19, 23, 29, 31, 37$	20	$41, 43, 47, 61$
14	$29, 31, 37, 41, 43$	21	$29, 31, 37, 41, 43, 47$
15	$23, 29, 31, 37, 43$	22	$37, 41, 43, 47, 53, 67$

TABLE 1. The pairs (d, p) for which we give an explicit point of order p over a number field of degree d.

4. Reduction to finitely many primes

In this section we show that for each $9 \leq d \leq 22$ it is enough to rule out the primes in Table 4 below. Throughout, $C = 2^{19}/325$. Note that Proposition 2.5 a) says that $\operatorname{gon}_{\mathbb{Q}} X_1(p) \geq 8(p^2-1)/C$.

4.1. **Oesterlé's bound and a version at the prime 2.** Let $\mathbb{T} \subseteq \operatorname{End} J_0(p)$ be the Hecke algebra generated over $\mathbb{Z}$ by the Hecke operators T_n, let $e \in H_1(X_0(p), \mathbb{Q})$ be the winding element, i.e. the class of the path from 0 to $i\infty$, let $I = \operatorname{Ann}_{\mathbb{T}}(e)$, and let

$$\pi : J_0(p) \to A = J_0(p)/IJ_0(p)$$

be the winding quotient. The group $A(\mathbb{Q})$ is finite [9, Theorem 2.2]. Following [9, Definition 6.5] we put

$$T'_n = \sum_{m|n} \mu(n/m) T_m, \qquad L_n = T'_{2n} - 2T'_n.$$

For an odd integer $M \geq 3$ let $\varepsilon_M(a) = 0$ if the representative of $a \in (\mathbb{Z}/M\mathbb{Z})^\times$ in $\{1, \ldots, M-1\}$ is less than $M/2$, and $\varepsilon_M(a) = 1$ otherwise.

Lemma 4.1. *Let $m \geq 1$, let ℓ be a prime, let $M \geq 3$ be odd, let $p > 2mM$ be a prime, and let u be such that $pu \equiv 1 \pmod{M}$. If the $m \times \varphi(M)$ matrix*

$$\big(\varepsilon_M(na) - \varepsilon_M(nu/a)\big)_{1 \leq n \leq m,\ a \in (\mathbb{Z}/M\mathbb{Z})^\times}$$

has rank m modulo ℓ, then $T_1 e, \ldots, T_m e$ are linearly independent in $\mathbb{T}e/\ell\mathbb{T}e$.

Proof. By [9, Lemma 6.7] the entries of the matrix are the intersection numbers of $L_n e \in H_1(X_0(p), \mathbb{Z})$ with the modular symbols $\{0, a/M\}$, so $L_1 e, \ldots, L_m e$ are linearly independent in $H_1(X_0(p), \mathbb{F}_\ell)$. The $\mathbb{Z}$-span of $L_1, \ldots, L_m$ is the same as that of $(T_2 - 3)T_1, \ldots, (T_2 - 3)T_m$, and $(T_2 - 3)\mathbb{T}e \subseteq H_1(X_0(p), \mathbb{Z})$ (see the proof of [9, Corollary 6.6]). So if $\sum \lambda_n T_n e \in \ell\mathbb{T}e$ with $\lambda_n \in \mathbb{Z}$, then $\sum \lambda_n (T_2 - 3) T_n e \in \ell H_1(X_0(p), \mathbb{Z})$, and hence all the λ_n are divisible by ℓ. □

Definition 4.2. Let ℓ be a prime and $d \geq 1$. We say that a prime p is *ℓ-admissible for d* if p divides $\#E(\mathbb{F}_{\ell^f})$ for some elliptic curve E over $\mathbb{F}_{\ell^f}$ with $f \leq d$, or if p divides $\ell^f - 1$ or $\ell^f + 1$ for some $f < d$.

Lemma 4.3. *Let $\ell \neq p$ be primes, let K be a number field of degree d, let E/K be an elliptic curve with a point P of order p, and let D_0 be the trace divisor of the point $(E, \langle P \rangle) \in X_0(p)(K)$. If p is not ℓ-admissible for d, then $\overline{D_0}$ (the reduction modulo ℓ) is d times one of the two cusps of $X_0(p)_{\mathbb{F}_\ell}$.*

Proof. Let v be a prime of K above ℓ, with residue field $\mathbb{F}_{\ell^f}$. If E had good reduction at v, then P would reduce to a point of order p, so E has bad reduction at v. The prime to ℓ torsion of $E(K_v)$ injects into the group of components when the reduction is additive, so as this group has order at most 4, the reduction is multiplicative. If P reduces to the identity component, then p divides $\ell^f - 1$ or $\ell^f + 1$, so $f = d$, v is the only prime above ℓ, and over the unramified extension of K_v over which E is a Tate curve, $\langle P \rangle$ is the canonical subgroup μ_p, so $(E, \langle P \rangle)$ reduces to the cusp ∞ at v. Otherwise $\langle P \rangle \neq \mu_p$, and $(E, \langle P \rangle)$ reduces to the cusp 0 at v. Hence either all the points in the support of D_0 reduce to 0, or there is just one prime above ℓ and they all reduce to ∞. □

The following proposition for odd ℓ is the argument of Oesterlé [9, Section 6] (see also [30, Theorem 4.18]). For $\ell = 2$ it is new. The extra factor 2 in $m = 2d$ compensates for the two ways in which reduction modulo 2 behaves worse than reduction modulo an odd prime: the kernel of reduction on rational torsion is only killed by 2, and the cotangent lattice of the Néron model of A may have index 2^r in its saturation.

Proposition 4.4. *Let ℓ be a prime, $d \geq 1$, and put $m = d$ if ℓ is odd and $m = 2d$ if $\ell = 2$. Let $p \neq \ell$ be a prime such that $T_1 e, \ldots, T_m e$ are linearly independent in $\mathbb{T}e/\ell\mathbb{T}e$. If $p \in S(d)$, then p is ℓ-admissible for d.*

Proof. Suppose the opposite. By Lemma 4.3, after applying the Atkin-Lehner involution if necessary, there is a non-cuspidal effective divisor D_0 of degree d on $X_0(p)$, defined over $\mathbb{Q}$, with $\overline{D_0} = d\infty$. For $k \geq 1$ consider the map

$$f_k : X_0(p)^{(k)} \to A, \qquad D \mapsto \pi[D - k\infty].$$

The point $f_d(D_0) \in A(\mathbb{Q})$ is torsion and it reduces to $f_d(d\infty) = 0$.

Suppose first that ℓ is odd. Then $f_d(D_0) = 0$ by Lemma 2.4 a), and f_d is a formal immersion at $d\infty$ modulo ℓ by [30, Theorem 4.18] (see [9, Proposition 6.3]), so $D_0 = d\infty$ by [9, Lemma 4.2], which is a contradiction.

Suppose now that $\ell = 2$. Then $2f_d(D_0) = f_{2d}(2D_0) = 0$ by Lemma 2.4 a). We use the following four facts.

a) Let $S = H^0(X_0(p)_{\mathbb{Z}_2}, \Omega^1)$, $M = S[I]$, and let $L = \pi^* H^0(A_{\mathbb{Z}_2}, \Omega^1) \subseteq M$, where $A_{\mathbb{Z}_2}$ is the Néron model of A. Then $2M \subseteq L$. This follows from a theorem of Raynaud [1, Appendix A, Theorem A.1], applied to the exact sequence $0 \to IJ_0(p) \to J_0(p) \to A \to 0$ of abelian varieties with good reduction over $\mathbb{Q}_2$, whose ramification index is $1 = 2 - 1$: the cokernel of the map of tangent spaces of the Néron models over $\mathbb{Z}_2$ is killed by 2 (compare [4, Section 4.4]). Dually, L has index killed by 2 in its saturation, which is M.

b) The pairing $\mathbb{T} \otimes \mathbb{Z}_2 \times S \to \mathbb{Z}_2$, $(t, \omega) \mapsto a_1(t\omega)$, is perfect (see [1, Lemma 2.1]), so $M \simeq \mathrm{Hom}(\mathbb{T}/I \otimes \mathbb{Z}_2, \mathbb{Z}_2)$. As $\mathbb{T}e \simeq \mathbb{T}/I$, our assumption implies that the map $M \to \mathbb{Z}_2^m$, $\omega \mapsto (a_1(\omega), \ldots, a_m(\omega))$, is surjective, where $\omega = \sum a_n(\omega) q^{n-1} dq$ is the q-expansion at ∞.

c) Let $s_1, \ldots, s_m$ be the elementary symmetric functions in the q-parameter at ∞, which are formal coordinates on $X_0(p)^{(m)}_{\mathbb{Z}_2}$ at $m\infty$. The cotangent map of f_m at $m\infty$ is $\omega \mapsto (a_1(\pi^*\omega), -a_2(\pi^*\omega), \ldots, (-1)^{m-1} a_m(\pi^*\omega))$ [30, Section 4.12], so by a) and b) its image contains $2\mathbb{Z}_2^m$. Hence there are formal functions $g_1, \ldots, g_m$ on $A_{\mathbb{Z}_2}$ at 0, vanishing at 0, such that

$$f_m^* g_i = 2s_i + Q_i(s_1, \ldots, s_m), \qquad Q_i \in (s_1, \ldots, s_m)^2.$$

d) As $\overline{D_0} = d\infty$, the polynomial $E_{D_0}(t) = \prod_j (1 - q_j t)$, where the q_j are the values of q at the points of D_0, has coefficients in $\mathbb{Z}_2$ and is congruent to 1 modulo 2. Hence $E_{2D_0}(t) = E_{D_0}(t)^2 \equiv 1 \pmod 4$, i.e. all the coordinates $s_i(2D_0)$ are in $4\mathbb{Z}_2$.

As $D_0 \neq d\infty$, not all $s_i(2D_0)$ are 0. Let $k \geq 2$ be the minimum of their valuations, attained at s_i. Then $2s_i(2D_0)$ has valuation $k+1$, while $Q_i(s(2D_0))$ has valuation at least $2k > k+1$. Hence $g_i(f_m(2D_0)) \neq 0$, which contradicts $f_m(2D_0) = f_{2d}(2D_0) = 0$. □

Remark 4.5. In particular, if $p > (2^{d/2}+1)^2$ and $T_1 e, \ldots, T_{2d} e$ are linearly independent in $\mathbb{T}e/2\mathbb{T}e$, then $p \notin S(d)$, which is the analogue of Oesterlé's bound with 3 replaced by 2. Using Parent's estimate [30, Proposition 1.9] instead of Lemma 4.1, the same argument should give $p \leq \max\{(2^{d/2}+1)^2, 65(4d)^6\}$ for all $p \in S(d)$ (see the note `dyadic/asymptotic_prime_bound.pdf`), but we do not need this.

For $d \in \{9, 10\}$ Oesterlé's bound $p \leq (3^{d/2}+1)^2$ [9, Theorem 6.1] is smaller than 10^5, and for the larger d we use Proposition 4.4.

Corollary 4.6. *Let $11 \leq d \leq 22$, let ℓ, M and $B(d)$ be as in Table 2, and let $p \in S(d)$. Then $p \leq B(d)$, and if $p > 10^5$, then p is one of the $N(d)$ primes listed in `large_primes.md`.*

Proof. Let $m = d$ if ℓ is odd and $m = 2d$ if $\ell = 2$. We compute that the matrices of Lemma 4.1 have rank m modulo ℓ for all $u \in (\mathbb{Z}/M\mathbb{Z})^\times$ ($d = 11$, $d = 12$, $d \geq 13$), so by Proposition 4.4 every $p \in S(d)$ with $p > 2mM$ is ℓ-admissible for d, for each ℓ in the table. As $2mM < 10^5$ in all cases, it remains to determine the admissible primes. An elliptic curve over $\mathbb{F}_{\ell^f}$ has at most $(\ell^{f/2}+1)^2$ points, and if its trace of Frobenius is divisible by ℓ, it is supersingular, so $j = 0$ for $\ell \in \{2, 3, 5\}$ and $j = 1728$ for $\ell = 7$, and we can list all the possible group orders. Note that an ordinary elliptic curve over $\mathbb{F}_{2^f}$ has a rational point of order 2, so a prime p dividing its order satisfies $2p \leq (2^{f/2}+1)^2$. For $d \geq 13$ this gives $p \leq B(d)$, and in all cases we compute the admissible primes above 10^5. □

d	ℓ	M	$2mM$	$B(d)$	$N(d)$
11	3, 5	53, 73	1166, 1606	177989	21
12	3, 5, 7	61, 59, 59	1464, 1416, 1416	532900	92
13	2	157	8164	8374	0
14	2	157	8792	16641	0
15	2	157	9420	33131	0
16	2	157	10048	66049	0
17	2	181	12308	66049	0
18	2	181	13032	263169	81
19	2	181	13756	525313	196
20	2	229	18320	525313	444
21	2	229	19236	1050024	850
22	2	229	20152	2099200	1488

TABLE 2. The primes ℓ and moduli M used in Lemma 4.1, the resulting bound $B(d)$ for the primes in $S(d)$, and the number $N(d)$ of admissible primes in $(10^5, B(d)]$. For $d \in \{11, 12\}$, $B(d)$ is Oesterlé's bound.

4.2. **The results of Derickx and Stoll.** We will use the following two results of Derickx and Stoll, which we state in our notation (see Lemma 2.4). For $d \geq 1$, let $S_{new}(d)$ be the set of primes p such that $X_1(p)$ has a non-cuspidal closed point of degree d, so that $S(d) = \bigcup_{d'|d} S_{new}(d')$.

Proposition 4.7 ([11, Corollary 7.5])**.** *Let $d \geq 1$ and $p \in S_{new}(d)$. Then one of the following holds:*

a) $p^2 \leq Cd + 1$,
b) *d is even and $p \in \{2d+1, 3d+1\}$,*
c) *$(p-1)/(2\operatorname{str}(p))$ divides d, and the image of the point on X_H, where $H \subseteq \Delta(p)$ is the subgroup of order $(p-1)/(2\operatorname{str}(p))$, has degree $2\operatorname{str}(p)d/(p-1)$.*

Proposition 4.8 ([11, Proposition 4.1])**.** *Let $a \in \Delta(p)$ and $\ell \in \{2, 3\}$, and put $n = 2\ell + 1$ if $a \in \{\ell, \ell^{-1}\}$ and $n = 2\ell + 2$ otherwise. Suppose that $(\langle a \rangle - 1)J_1(p)(\mathbb{Q})$ is finite and, if $\ell = 2$, that its 2-primary part is killed by A_2. Let D be an effective divisor of degree d on $X_1(p)$, defined over $\mathbb{Q}$, with no cusps in its support. If $nd < \operatorname{gon}_{\mathbb{Q}} X_1(p)$, then $\langle a \rangle D = D$. The same holds with $X_1(p)$ replaced by X_H for a subgroup $H \subseteq \Delta(p)$ and $a \in \Delta(p)/H$.*

By Lemma 2.4 b) the assumption for $\ell = 2$ holds if $p \leq 157$. For larger p we use the following lemma.

Lemma 4.9. *Let $h \in \Delta(p)$ be such that $(h-1)J_1(p)(\mathbb{Q})$ is finite, and suppose that there is an endomorphism β of $J_1(p)$ which kills $J_1(p)(\mathbb{Q})_{\text{tors}}$ and restricts to an isogeny of odd degree on $B = (h-1)J_1(p)$. Then $(h-1)J_1(p)(\mathbb{Q})$ has odd order.*

Proof. The group $(h-1)J_1(p)(\mathbb{Q})$ is contained in $B(\mathbb{Q})$ and is killed by β. □

We will apply this lemma with $\beta = A_3$, or a combination of several A_q with q odd. To show that A_3 has odd degree on B we compute its determinant on the rational homology of B, which is the sum of the components of $H_1(X_1(p), \mathbb{Q})$ belonging to the characters χ with $\chi(h) \neq 1$, and on each of them the determinant is an element of the field of values of χ whose norm is odd.

4.3. **Primes above 10^5.**

Lemma 4.10. *Let p be a prime with $p^2 > Cd + 1$ and $p \notin \{2d'+1, 3d'+1\}$ for all $d' \mid d$, let $m = (p-1)/2$, and let r be a prime with $v_r(m) > v_r(d)$. If no even character modulo p whose order n satisfies $v_r(n) \geq v_r(m) - v_r(d)$ is strange, then $p \notin S(d)$.*

Proof. The strange characters generate a cyclic group of order $\operatorname{str}(p)$, and by assumption $v_r(\operatorname{str}(p)) < v_r(m) - v_r(d)$. Hence $v_r(m/\operatorname{str}(p)) > v_r(d)$, so $m/\operatorname{str}(p)$ does not divide any divisor d' of d, and the result follows from Proposition 4.7. □

To show that a character is not strange we use the method of [11, Section 5], made rigorous as follows.

Lemma 4.11. *Let χ be a non-trivial even character modulo p of order n, and let $q \nmid 6p(p-1)$ be a prime with $q \equiv 1 \pmod{n}$, so χ takes values in $\mathbb{F}_q$. Let V be the plus part of the space of modular symbols of weight 2, level p and character χ over $\mathbb{F}_q$, let $V_0 \subseteq V$ be its cuspidal subspace, of dimension k, and let $e' = (T_2 - 1 - 2\chi(2))(T_2 - 2 - \chi(2))\{0, \infty\}$, which lies in V_0. If $e', T_2e', \ldots, T_2^{k-1}e'$ are linearly independent, then χ (and every character of order n) is not strange.*

Proof. The space V and the action of T_2 are the reductions of their analogues over the ring of integers of $\mathbb{Q}(\chi)$ localized at a prime above q, so the corresponding vectors in characteristic 0 are linearly independent as well. Hence e' generates the cuspidal part as a module over the Hecke algebra, so its projection to the eigenspace of every newform f with character χ is non-zero. This projection is a non-zero multiple of $L(f,1)$, so $L(f,1) \neq 0$. The characters of order n are Galois conjugate. □

In practice we take a random linear functional λ on V_0 and check that the sequence $\lambda(T_2^i e')$, $0 \leq i < 2k$, has linear complexity k, i.e. that it does not satisfy a linear recurrence of length less than k, which implies the assumption of Lemma 4.11. We certify the linear complexity by a rational function A/C with $C(0) = 1$, $\deg C \leq k$, $\deg A < k$, $\max(\deg C, \deg A + 1) = k$, $\gcd(A, C) = 1$, and $C \cdot \sum_i \lambda(T_2^i e')x^i \equiv A \pmod{x^{2k}}$. For p up to $2.1 \cdot 10^6$ the dimension k is close to $p/12$, so these are large sparse computations, for which we wrote our own implementation (see Section 7).

Proposition 4.12. *Let $9 \leq d \leq 22$ and $p > 10^5$. Then $p \notin S(d)$.*

Proof. For $d \leq 10$ this follows from Oesterlé's bound, and for $13 \leq d \leq 17$ from Corollary 4.6. For the other d, let p be one of the $N(d)$ primes of Corollary 4.6. For each of them we choose r as in Lemma 4.10 and show, using Lemma 4.11, that no character of the required orders is strange. This requires 73, 656, 615, 1147, 3020, 5289 and 10065 character orders for $d = 11, 12, 18, 19, 20, 21, 22$, respectively, many of which are shared between the degrees ($d = 11$, $d = 12$, $d = 18$, $d \geq 19$). The result follows from Lemma 4.10. □

4.4. Primes below 10^5.

Proposition 4.13. *Let $9 \leq d \leq 22$, and let $p < 10^5$ be a prime which is neither one of the primes listed for d in Theorem 1.1 nor one of the primes listed for d in Table 4. Then $p \notin S(d)$.*

Proof. Let $p \in S_{new}(d')$ for some $d' \mid d$. If $p^2 > Cd + 1$, then by Proposition 4.7 and [11, Table 2], p is one of 181 (for $d = 15$), 193 (for $d = 16$), or 409 and 1021 (for $d = 17$). We check this for all $p < 10^5$, using that the possible exceptions b) of Proposition 4.7 are in $S(d)$. For $p \in \{181, 409, 1021\}$ we have $(p-1)/(2\,\mathrm{str}(p)) = d$, so Proposition 4.7 c) gives a non-cuspidal rational point on $X_0(p)$, which does not exist by Mazur's theorem [23]. The prime 193 is dealt with below.

The remaining primes satisfy $p^2 \leq Cd + 1$, and they are either in Table 4, or in Table 3, or one of the primes treated below. For (p, a, ℓ, n) and d as in Table 3, the assumptions of Proposition 4.8 hold: $(\langle a \rangle - 1)J_1(p)(\mathbb{Q})$ is finite by Lemma 2.3 (the only strange p in the table is 181, and a is trivial on the characters of order 6 modulo 181), the gonality bound holds by Proposition 2.5 a), and the 2-primary assumption holds by Lemma 2.4 b) for $p \leq 157$ and by Lemma 4.9 for the larger p with $\ell = 2$ (see below). Hence the trace divisor D is invariant under $\langle a \rangle$. As $p > 67$, Lemma 2.2 shows that the orbits of $\langle a \rangle$ on its support have length $\mathrm{ord}(a)$, which is listed in Table 3 and does not divide d'. This is a contradiction.

For $p \in \{163, 167, 173\}$ and $\ell = 2$ we compute ($p = 167$, $p = 173$) that A_3 restricts to an isogeny of odd degree on $(\langle a \rangle - 1)J_1(p)$, so Lemma 4.9 applies. For $p = 181$ the operator A_3 has even degree on some components, and we use instead $\beta = EA_{11} + (45 - E)A_3$, where $u = \langle 2 \rangle^2$ (of order 45) and $E = 5N_{u^5} - N_u$, N_g denoting the sum of the powers of g. On the components of $H_1(X_1(181), \mathbb{Q})$ belonging to characters of order 5 and 10, E acts as 45, and on the other components of $(\langle a \rangle - 1)J_1(181)$ it acts as 0. We compute that A_{11}, respectively A_3, has odd determinant on these components, so β is as in Lemma 4.9.

It remains to deal with $p = 127$ for $d \in \{14, 16\}$ and with $p = 193$ for $d = 16$.

Let $p = 127$, so $\mathrm{ord}(\langle 2\rangle) = 7$. For $d = 16$, Proposition 2.5 a) gives $\mathrm{gon}_{\mathbb{Q}} X_1(127) \geq 84 > 5{\cdot}16$, so Proposition 4.8 with $(a, \ell, n) = (2, 2, 5)$ applies, and D is $\langle 2\rangle$-invariant, which is impossible as $7 \nmid 16$. For $d = 14$, Proposition 4.8 still applies, so D is $\langle 2\rangle$-invariant, and by Lemma 2.2 $D = \pi^* D'$ for an effective divisor D' of degree 2 on $Y = X_1(127)/\langle 2\rangle$, defined over $\mathbb{Q}$, with no cusps in its support. We compute that $X_1(127)_{\mathbb{F}_2}$ has 48 supersingular closed points of degree 14, each fixed by $\langle 2\rangle$, and these map to 48 distinct closed points of degree 2 on $Y_{\mathbb{F}_2}$. Hence $\#Y(\mathbb{F}_4) \geq 96$ and $\mathrm{gon}_{\mathbb{Q}} Y \geq \mathrm{gon}_{\mathbb{F}_4} Y \geq 96/5$, so $\mathrm{gon}_{\mathbb{Q}} Y \geq 20$. The image of $\langle 3\rangle$ in $\Delta(127)/\langle 2\rangle$ has order 9, and Proposition 4.8 on Y with $(a, \ell, n) = (3, 3, 7)$ and $7 \cdot 2 < 20$ shows that D' is a sum of $\langle 3\rangle$-orbits. These have length 9 or 3, which is a contradiction.

Let $p = 193$ and $d = 16$. Then $\mathrm{str}(193) = 12$, so by Proposition 4.7 the image of the point on $Y = X_H$, where H has order 8, is a rational point (if $d' = 8$), which is impossible by Mazur's theorem, or a non-cuspidal point y of degree 2 (if $d' = 16$). If y had non-cuspidal reduction modulo 2, then a point above it on $X_1(193)_{\mathbb{F}_2}$ would have degree at most 16 and j-invariant in $\mathbb{F}_4$, and we compute that there are no such points. The non-rational cusps of $Y_{\mathbb{F}_2}$ have degree 6, so the trace divisor of y reduces to a sum of two rational cusps of Y. We compute an element α of the Hecke algebra of J_Y with $\alpha J_Y(\mathbb{Q})$ finite, and check that the map $Y^{(2)} \to J_Y$ given by $t = \alpha A_3$ is a formal immersion at all these divisors, which contradicts Lemma 2.9. □

p	a	ℓ	n	d	$\mathrm{ord}(a)$
103	2	2	5	9, 10	51
107	2	2	5	9–11	53
109	2	2	5	9–11	18
113	2	2	5	9–12	14
127	2	2	5	10–13, 15	7
131	2	2	5	11–17	65
137	2	2	5	12–18	34
139	2	2	5	12–19	69
149	2	2	5	14–22	74
151	6	2	6	15	75
151	2	2	5	16–22	15
157	2	2	5	16–22	26
163	3	3	7	17, 18	81
163	8	2	6	19–21	27
167	3	3	7	18, 19	83
167	2	2	5	20–22	83
173	3	3	7	19–21	86
173	4	2	6	22	43
179	3	3	7	20–22	89
181	64	2	6	21, 22	15

TABLE 3. The applications of Proposition 4.8. Here $\mathrm{ord}(a)$ is the order of a in $\Delta(p)$.

d	primes p treated in Sections 5 and 6
9	$37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101$
10	$41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101$
11	$37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103$
12	$41, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109$
13	$41, 43, 47, 53, \ldots, 113$
14	$47, 53, \ldots, 113$
15	$41, 47, 53, \ldots, 113$
16	$41, 43, 47, 53, \ldots, 113$
17	$47, 53, \ldots, 113, 127$
18	$47, 53, \ldots, 113, 127, 131$
19	$47, 53, \ldots, 113, 127, 131, 137$
20	$53, 59, 67, 71, \ldots, 113, 127, 131, 137, 139$
21	$53, 59, 61, 67, \ldots, 113, 127, 131, 137, 139$
22	$59, 61, 71, 73, \ldots, 113, 127, 131, 137, 139, 163$

TABLE 4. The primes that remain after Propositions 4.12 and 4.13. Here $53, \ldots, 113$ denotes all the primes from 53 to 113.

5. The local method at the prime 2

In this section and the next we deal with the pairs (d, p) of Table 4. We follow [9, Section 8], and we reduce the trace divisor modulo 2. Throughout, (d, p) is one of these pairs, D is the trace divisor of a point of order p over a number field of degree d, and we write

$$\overline{D} = Z + C',$$

where C' is supported on the $\mathbb{F}_2$-rational cusps of $X_1(p)_{\mathbb{F}_2}$ and Z has no such points in its support. Let C be the rational cuspidal divisor lifting C', and $c = \deg C$.

5.1. **The census.** For each $f \leq d$ we enumerate all elliptic curves E over $\mathbb{F}_{2^f}$ up to isomorphism, and for each of them all the points of order p of $E(\mathbb{F}_{2^f})$ up to the action of $\mathrm{Aut}(E)$ (all the cyclic subgroups of order p when $E[p] \subseteq E(\mathbb{F}_{2^f})$). This gives all the non-cuspidal closed points of $X_1(p)_{\mathbb{F}_2}$ of degree at most d, together with the action of $\Delta(p)$ on them. For $d \geq 20$ we check the number of these points independently, using class numbers for the ordinary points and the description of the supersingular ones by quaternions.

A non-cuspidal closed point of $X_1(p)_{\mathbb{F}_2}$ is *supersingular* if its j-invariant is 0, and *ordinary* otherwise. We write O_k, SS_k and NC_k for an ordinary closed point, a supersingular closed point and a non-rational cusp of degree k, and C_c for an effective divisor of degree c supported on the rational cusps. The *type* of $\overline{D}$ is the corresponding formal sum, e.g. $O_8 + SS_7 + C_7$. The supersingular points and the non-rational cusps of degree at most 22 at the primes we need are listed in Table 5. An ordinary elliptic curve over $\mathbb{F}_{2^f}$ has a rational point of order 2, so if it has a point of order p, then $2p \leq (2^{f/2} + 1)^2$ by the Hasse bound. Hence ordinary closed points have degree at least 6, and at least 7 for $p \geq 41$, so Z contains at most three ordinary points.

p	supersingular points	non-rational cusps
37	SS_9 (1)	
41	SS_5 (2)	NC_{10} (2)
43	SS_7 (3), SS_{14}	NC_7 (3)
53	SS_{13} (1)	
61	SS_{15} (1), SS_{20}	
73	SS_6 (4), SS_{18}	NC_9 (4)
89	SS_{22} (15)	NC_{11} (4)
97	SS_{16} (2)	
109	SS_9 (3), SS_{12} (3)	NC_{18} (3)
113	SS_7 (4)	NC_{14} (4)
127	SS_{14} (48)	NC_7 (9)
137	SS_{17}	

TABLE 5. The supersingular points and non-rational cusps of degree at most 22 on $X_1(p)_{\mathbb{F}_2}$, for the primes p of Table 4 at which there are any, with the number of closed points of each degree in brackets where we use it. Only the degrees that occur in the relevant range of d are listed.

For each (d, p) we enumerate all the possible types of $\overline{D}$, and for each type all the possible Z up to the action of $\Delta(p)$, which we may use as $\Delta(p)$ commutes with reduction and with all the operators we apply. For $d = 22$ there are 514 types at the 18 primes, and we checked the list by a second, independent enumeration.

5.2. **Reductions supported on the rational cusps.** Suppose that $Z = 0$. We use Lemma 2.9 on $X_1(p)$ or on an intermediate curve X_H, with $t = \alpha A_q$ for an element α of the Hecke algebra with $\alpha J_H(\mathbb{Q})$ finite and an odd prime q, or with a product of such maps, and we check the criterion of [9, Proposition 5.3] for all the multiplicity vectors (d_k) of total degree d. Using the action of $\Delta(p)$ we may assume that $d_0 = \max_k d_k$. When the number of vectors is too large, we instead compute all the linear relations modulo 2 between the vectors of q-expansion coefficients and check that each of them involves more than d coefficients in total.

Proposition 5.1. *For all the pairs (d, p) in Table 4 except $p = 53$ with $d \geq 19$, $p = 101$ with $d \geq 20$, $p = 103$ with $d \geq 17$ and $p = 131$ with $d \geq 18$, the map of Lemma 2.9 is a formal immersion at every sum of rational cusps of degree d. In particular $Z \neq 0$.*

The data of these computations (the subgroups H, the maps t, and the numbers of vectors) are in the files `formal_immersion.json` in the repository. In several cases (for example at $p = 71$ and $p = 127$ for $d \geq 21$) the computation on the intermediate curve we first tried fails, and we work on $X_1(p)$ itself. The four exceptional cases are treated in Section 6.

5.3. **The Hecke obstruction.** For $Z \neq 0$ we use the following form of [9, Lemma 8.6].

Lemma 5.2. *Let $\alpha \in \mathbb{Z}[\Delta(p)]$ be such that $\alpha J_1(p)(\mathbb{Q})$ is finite, and let $q \neq p$ be an odd prime. Then the divisor $\alpha A_q Z$ on $X_1(p)_{\mathbb{F}_2}$ is principal.*

Proof. The class $\alpha[D - dc_0]$ is torsion, so $A_q\alpha[D - dc_0] = 0$ by Lemma 2.4 b), and hence its reduction $[\alpha A_q(Z + C' - dc_0)]$ is 0. As α maps rational cusps to sums of rational cusps, and A_q kills each rational cusp, $\alpha A_q(C' - dc_0) = 0$ as a divisor. □

We use α as in Table 6, which have $\alpha J_1(p)(\mathbb{Q})$ finite by Lemma 2.3, and almost always $q = 3$. To compute $A_3 Z$ we use the 3-isogenies of the elliptic curves representing the points of Z, and we show that $\alpha A_3 Z$ is not principal using Lemma 2.8 with $\mathcal{L} = \mathcal{M} = K$, or $2K$ when the degree of the divisor is too large (at $p \leq 53$, and at 61, 67 and 97 for some d). If a part Z_1 of Z is invariant under a subgroup $H \subseteq \Delta(p)$, we replace α by $(h-1)\alpha$ with $h \in H$, which kills Z_1. For example, at $p = 41$ we use $\alpha = \langle a \rangle^2 - 1$, where $\langle a \rangle$ generates $\Delta(41)$, for $Z = SS_5 + O_k$, as $\langle a \rangle^2$ fixes both closed points of degree 5.

p	α
$41, 47, 59, 71$	1
61	$\sigma^6 - 1$
97	$\sigma^{12} - 1$, $(\sigma - 1)\Phi_{12}(\sigma)$ or $(\sigma - 1)(\sigma^6 + 1)$
101	$(\sigma - 1)(\sigma^5 + 1)$
other p	$\sigma - 1$

TABLE 6. The elements α used in Lemma 5.2, where σ is a generator of $\Delta(p)$ and Φ_{12} is the 12-th cyclotomic polynomial. They kill the trivial character and the strange characters of orders 6, 12 and 10 modulo 61, 97 and 101.

Proposition 5.3. *Let (d, p) be a pair from Table 4. Then Z is one of the divisors listed for p in Table 7.*

Proof. By Proposition 5.1, $Z \neq 0$ (apart from the four exceptional cases, which are listed in the table). For all the other possibilities for Z given by the census, we compute that $\alpha A_3 Z$ (or, in a few cases, $\alpha A_5 Z$ or $\alpha A_7 Z$) is not principal (pairs and triples, mixed supports), which contradicts Lemma 5.2. Each of these computations depends only on Z, so a computation made for one d applies to all larger d as well. For example, at $p = 43$, if $Z = C$ is one of the three non-rational cusps, then up to sign and a diamond operator $(\langle 3 \rangle - 1)A_3 Z$ is $2(2\langle 3 \rangle C - \langle 9 \rangle C - C)$, and we compute that this divisor is not principal, which excludes $Z = NC_7$ in all degrees. □

These computations involve a very large number of divisors, on curves of genus up to 1027. For example, for $d = 22$ alone there are 76185 new orbits under $\Delta(p)$ of ordinary closed points of degree 22. Note that the test cannot succeed if Z is the reduction of a divisor D_1 defined over $\mathbb{Q}$, for example of a point of smaller degree on $X_1(p)$ plus some rational cusps. In fact, some of the ordinary points in Table 7 are reductions of points of degree 10, 12 and 14 on $X_1(37)$, $X_1(41)$ and $X_1(43)$, and the others have a large stabilizer in $\Delta(p)$.

p	Z	d
37	O_6, O_9, SS_9	9, 11
37	two orbits of O_{10}	11
41	O_{14}	15, 16
43	O_{12}	13, 16
43	O_{14}, O_{15}, SS_{14}, $2SS_7$, $2NC_7$, $SS_7 + NC_7$	16
53	SS_{13}	13–21
53	0	19–21
61	SS_{15}	15–19, 21, 22
61	SS_{20}, O_{20}	21, 22
73	sums of SS_6, NC_9 and SS_{18}	9–22
89	NC_{11}, $2NC_{11}$, SS_{22}	11–22
97	SS_{16}	16–22
101	0	20–22
103	0	17–22
109	SS_9, NC_{18}, $2SS_9$, $SS_9 + SS_{12}$	12–22
113	$2SS_7$, $3SS_7$, NC_{14}, $SS_7 + NC_{14}$	14–22
113	O_8, $2O_8$, $O_8 + SS_7$, $O_8 + NC_{14}$	13–22
127	sums of NC_7 and SS_{14}	17–22
127	O_9, $2O_9$, $O_9 + NC_7$	17–22
131	0	18–22
137	SS_{17}, O_{17}	19–22

TABLE 7. The reductions Z that are not excluded by Propositions 5.1 and 5.3, and the degrees d in which they have to be considered. Here O_k denotes the exceptional ordinary closed points described in Section 6, and $Z = 0$ means that $\overline{D}$ is supported on the rational cusps.

6. The remaining reductions

In this section we deal with the reductions in Table 7, keeping the notation of Section 5. Recall that $\overline{D} = Z + C'$, that C is the rational cuspidal divisor lifting C', and $c = \deg C = d - \deg Z$. Most of the cases are dealt with by Lemma 2.7, in which case we give the element h, the reason why $y = 0$ (if we use it), and the relevant inequality for the gonality, which always comes from Proposition 2.5. All the statements that h fixes a closed point of $X_1(p)_{\mathbb{F}_2}$ are checked on the elliptic curves representing these points. When we say that D *is a pullback from* X_H, we mean that $D = \pi^* D'$, where $\pi : X_1(p) \to X_H$ is the quotient map and D' is an effective divisor on X_H defined over $\mathbb{Q}$, with no cusps in its support. By Lemma 2.2, this is the case if D is invariant under H and $d < (p-1)/3$.

6.1. The prime 37.

Proposition 6.1. *We have* $37 \notin S(9)$ *and* $37 \notin S(11)$.

Proof. We compute that the common kernel of A_3 and A_5 on $H_1(X_1(37), \mathbb{Z}) \otimes \mathbb{F}_2$, which contains $J_1(37)(\mathbb{Q})[2]$ by Lemma 2.4 b), is 0. So $y = 0$ in Lemma 2.7, and we have $\operatorname{gon}_{\mathbb{Q}} X_1(37) = 18$. The exceptional reductions are the following orbits.

Suppose first that $Z = O_6$, one of 3 closed points of degree 6 fixed by $h = \langle 8 \rangle$, of order 6. Then $c = d - 6 \in \{3, 5\}$, and $18 > d + c$, so $6 \mid c$ by Lemma 2.7, which is a contradiction.

Suppose now that Z is one of the 2 ordinary closed points of degree 9, or the supersingular closed point of degree 9, all of which are fixed by $h = \langle 4 \rangle$, of order 9. Then $c \in \{0, 2\}$ and $18 > d + c$, so $hD = D$ and $9 \mid c$. Hence $c = 0$, $d = 9$, and by Lemma 2.2 the support of D is a single orbit of $\langle 4 \rangle$, so the image of D on $X_0(37)$ is a non-cuspidal rational point. The corresponding j-invariants are $-7 \cdot 11^3$ and $-7 \cdot 137^3 \cdot 2083^3$ [24, Section 5], those of the curves 1225.b2 and 1225.b1. As E is a quadratic twist of one of these curves over K, the x-coordinate of P is a root of the 37-division polynomial of one of them which lies in K. But we compute that the irreducible factors of these division polynomials have degrees $6, 6, 6, 666$ and $18, 222, 222, 222$, none of which divides 9.

Suppose finally that $d = 11$ and $Z = O_{10}$, so $c = 1$. For one of the two orbits, Z is fixed by $h = \langle 6 \rangle$, of order 2, and as $18 > 12$, Lemma 2.7 gives $2 \mid 1$. The other orbit consists of the reductions of the closed points of degree 10 in the supports of the trace S of the point of order 37 of Section 3 and of its images under $\Delta(37)$ (we check this). Replacing S by such an image we can assume $\overline{S} = Z$, so D and $S + C$ have the same reduction. We compute that for $t = \langle 2 \rangle - 1$ the map f of Lemma 2.9 is a formal immersion at $Z + \overline{c_1}$ for each of the 18 rational cusps c_1. As $J_1(37)(\mathbb{Q})[2] = 0$, reduction is injective on $tJ_1(37)(\mathbb{Q})$, so $D = S + C$ by Lemma 2.9, which is a contradiction as D contains no cusps. □

6.2. **The primes** 41 **and** 43**.**

Proposition 6.2. *We have* $41 \notin S(15)$ *and* $41 \notin S(16)$ *if* $Z = O_{14}$.

Proof. Here O_{14} is one of the 10 closed points of degree 14 which are the reductions of the closed points of degree 14 in the supports of the trace S of an explicit point of degree 14 on $X_1(41)$ and of its images under $\Delta(41)$, as we check. As in the proof of Proposition 6.1 we can assume that $\overline{S} = Z$. As $J_1(41)(\mathbb{Q})$ is finite, the map $t = (A_3, A_5, A_7) : J_1(41) \to J_1(41)^3$ kills $J_1(41)(\mathbb{Q})$, and we compute that f is a formal immersion at $Z + C'$ for all the 20 (for $d = 15$) and 210 (for $d = 16$, code) possible C'. So $D = S + C$ by Lemma 2.9, which is a contradiction. □

For $p = 43$ and $d = 16$ we need the following lemma.

Lemma 6.3. *Reduction modulo* 2 *is injective on* $J_1(43)(\mathbb{Q})_{\mathrm{tors}}$.

Proof. Let $Y = X_1(43)/\langle 2 \rangle$, where $\langle 2 \rangle$ has order 7, and let $f : X_1(43) \to Y$ be the quotient map. The curve Y has genus 9, and we compute that J_Y contains an elliptic curve A isogenous to 43.a1, whose isogeny class consists of one curve with trivial rational torsion, so $A(\mathbb{Q})[2] = 0$, that $B = J_Y/A$ has rank 0, and that the numbers of points of B over $\mathbb{F}_3$, $\mathbb{F}_5$ and $\mathbb{F}_{23}$ have greatest common divisor $532 = 4 \cdot 133$. Hence $J_Y(\mathbb{Q})[2]$ injects into $B(\mathbb{Q})_{\mathrm{tors}}$, which injects into $B(\mathbb{F}_3)$, $B(\mathbb{F}_5)$ and $B(\mathbb{F}_{23})$, so $\#J_Y(\mathbb{Q})[2] \leq 4$. Let C_1, C_2, C_3 be the rational cusps of Y. The divisors $38(C_2 - C_1)$ and $38(C_3 - C_1)$ are principal, and we compute that the reductions of $x_1 = 19[C_2 - C_1]$, $x_2 = 19[C_3 - C_1]$ and $x_1 + x_2$ are non-zero. So $J_Y(\mathbb{Q})[2] = \{0, x_1, x_2, x_1 + x_2\}$, and reduction is injective on it.

Now let $x \in J_1(43)(\mathbb{Q})_{\mathrm{tors}}$ with $\overline{x} = 0$. Then $2x = 0$, so $f_* x \in J_Y(\mathbb{Q})[2]$ has reduction 0, and hence $f_* x = 0$. We compute that $\langle 2 \rangle$ acts trivially on the common kernel of A_3, A_5 and A_7 on $H_1(X_1(43), \mathbb{Z}) \otimes \mathbb{F}_2$, which contains x. Hence $0 = f^* f_* x = \sum_{i=0}^{6} \langle 2 \rangle^i x = 7x = x$. □

Proposition 6.4. *We have* $43 \notin S(13)$ *and* $43 \notin S(16)$ *if* Z *is one of the divisors listed for* 43 *in Table 7.*

Proof. The 7 closed points O_{12} are the reductions of the closed points of degree 12 in the supports of the trace S of the point of order 43 of Section 3 and its images under $\Delta(43)$, as

we check, and the 21 closed points O_{14} are, in the same way, the reductions of the trace of an explicit point of degree 14 on $X_1(43)$ from the tables of [38] and its images (check). For $d = 13$ and $Z = O_{12}$ we compute that f is a formal immersion at $Z + \overline{c_1}$ for all the 21 rational cusps c_1, with $t = (\langle 3\rangle^{-1} - 1)\langle 11\rangle^{-1}A_{11}$, which kills $J_1(43)(\mathbb{Q})$, and for $d = 16$ and $Z = O_{14}$ we compute that f is a formal immersion at all the 231 divisors $Z + C'$, with t combining several maps αA_q, $q \in \{3, 5, 7\}$, with $\alpha J_1(43)(\mathbb{Q})$ finite. As in the proof of Proposition 6.1, this is a contradiction.

The remaining cases are for $d = 16$. By Lemma 6.3, $y = 0$ in Lemma 2.7, and $\operatorname{gon}_{\mathbb{Q}} X_1(43) = 24$. If $Z = O_{12}$ or Z is one of the two orbits O_{15}, then Z is fixed by $h = \langle 37\rangle$, of order 3, and $c \in \{4, 1\}$, so as $24 > 16 + c$, Lemma 2.7 gives $3 \mid c$, which is a contradiction. If $Z \in \{SS_{14}, 2SS_7, 2NC_7, SS_7 + NC_7\}$, then Z is fixed by $h = \langle 2\rangle$, of order 7, and $c = 2$, so as $24 > 18$ we get $7 \mid 2$, which is again a contradiction. □

6.3. **The primes 53 and 61.**

Proposition 6.5. *We have $53 \notin S(d)$ for $13 \le d \le 21$ if $Z = SS_{13}$ or $Z = 0$.*

Proof. There is a unique supersingular closed point SS_{13} of degree 13, and $h = \langle 4\rangle$, of order 13, acts simply transitively on its geometric points. We compute that $J_1(53)(\mathbb{Q})[2] = 0$, so $y = 0$ in Lemma 2.7.

Suppose $Z = SS_{13}$, so $c = d - 13$. If $d = 13$, then as $\operatorname{gon}_{\mathbb{Q}} X_1(53) > 13$, Lemma 2.7 gives $hD = D$, and by Lemma 2.2 D is a single orbit of $\langle 4\rangle$, so its image on $X_0(53)$ is a non-cuspidal rational point, which does not exist [23]. If $14 \le d \le 16$, then $d + c \le 19 < 20 \le \operatorname{gon}_{\mathbb{Q}} X_1(53)$, and Lemma 2.7 gives $13 \mid c$, which is a contradiction. For $17 \le d \le 21$ we use Lemma 2.10 with $L = C + \pi^*Y$, where $Y = X_1(53)/\langle 4\rangle$, and $t = ((h-1)A_3, (h-1)A_5)$, which kills $J_1(53)(\mathbb{Q})$ and π^*J_Y. We compute that the rank condition holds for all the possible C', so D lies on L and contains $C \neq 0$, which is a contradiction.

Suppose now that $Z = 0$ and $19 \le d \le 21$. For $d = 19$ the criterion of Proposition 5.1 on X_H with H of order 2 fails at only 7 multiplicity vectors, and at all the 20444 divisors on $X_1(53)$ lying above them we compute that f is a formal immersion, with $t = ((\langle 10\rangle - 1)A_3, (\langle 10\rangle - 1)A_5)$. For $d \in \{20, 21\}$ we work on $X_1(53)$ with $t = ((h-1)A_3, (h-1)A_5)$, and we compute that f is a formal immersion at every sum of rational cusps of degree d that does not contain a full orbit of $\langle 4\rangle$, while at the others the rank condition of Lemma 2.10 holds with $L = R + \pi^*Y$, where $R \neq 0$ is the rest of the divisor. In both cases D contains a cusp, which is a contradiction. □

Proposition 6.6. *We have $61 \notin S(d)$ for $d \in \{15, \dots, 19, 21, 22\}$ if $Z \in \{SS_{15}, SS_{20}, O_{20}\}$.*

Proof. There is a unique supersingular closed point SS_{15} of degree 15, and it is fixed by all of $\Delta(61)$. Let $h = \langle 3\rangle = \langle 2\rangle^6$, of order 5. As the strange characters modulo 61 have order 6, $(h-1)J_1(61)(\mathbb{Q})$ is finite, and we compute that $J_1(61)(\mathbb{Q})[2] = 0$, so $y = 0$ in Lemma 2.7. Note that $\operatorname{gon}_{\mathbb{Q}} X_1(61) > 18$.

Suppose $Z = SS_{15}$. If $d = 16$, then $c = 1$, and Lemma 2.7 gives $5 \mid 1$. If $d \in \{17, 18, 19, 21, 22\}$, we use Lemma 2.10 with $L = C + \pi^*Y^{(3)}$, where $Y = X_1(61)/\langle h\rangle$, and $t = ((h-1)A_3, (h-1)A_5)$, and we compute that the rank condition holds for all possible C', so D contains $C \neq 0$, which is impossible. In the same way, for $d \in \{21, 22\}$ and $Z \in \{SS_{20}, O_{20}\}$, where O_{20} is an orbit of 6 closed points which are pullbacks from Y, we use $L = C + \pi^*Y^{(4)}$ (SS_{20}, O_{20}).

It remains to deal with $d = 15$ and $Z = SS_{15}$, so $c = 0$. By Lemma 2.7 $hD = D$, so by Lemma 2.2 D is a pullback from Y, $D = \pi^*D_Y$, and D_Y is a closed point of degree 3. Let $W = X_1(61)/\langle 4\rangle$, where $\langle 4\rangle$ has order 15. Then W has genus 8, the image s of $\langle 2\rangle$ is an involution of W, and $W/\langle s\rangle = X_0(61)$. The image D_W of D_Y on W is a closed point

of degree 3, as a rational point would map to a non-cuspidal rational point on $X_0(61)$. Its reduction is fixed by s. The Jacobian J_W corresponds to the characters modulo 61 of order at most 2, so $(s-1)J_W(\mathbb{Q})$ is finite by Kato's theorem as in Lemma 2.3, and $J_W(\mathbb{Q})[2] = 0$ since $J_1(61)(\mathbb{Q})[2] = 0$ and the pullback to $X_1(61)$ is injective on 2-torsion (its composition with the pushforward is multiplication by 15). Hence, as in Lemma 2.7, $sD_W \sim D_W$, and as $\operatorname{gon}_{\mathbb{Q}} W \geq 4$ by Lemma 6.7 below, $sD_W = D_W$. As 2 and 3 are coprime, s fixes each of the three geometric points of D_W. If w is one of them and x is a point of D above it, then $\langle 2\rangle x = gx$ for some $g \in \langle 4\rangle$, so x is fixed by $\langle 2\rangle g^{-1} \neq 1$. This contradicts Lemma 2.1, as x has degree $15 < 20$. □

Lemma 6.7. *The curve $W = X_1(61)/\langle 4\rangle$ has no map of degree 2 or 3 to $\mathbb{P}^1$ defined over $\mathbb{Q}$.*

Proof. The curve $X_0(61)$ has genus 4 and no map of degree at most 3 to $\mathbb{P}^1$ over $\mathbb{Q}$: its canonical model is a complete intersection of a quadric and a cubic in $\mathbb{P}^3$, and the discriminant of the quadric is not a square, so the two g^1_3's are conjugate over $\mathbb{Q}(i)$ (see [14, 15] and our check). Suppose that $u : W \to \mathbb{P}^1$ has degree $e \in \{2, 3\}$. As $g(W) = 8 > (e-1)^2$, the Castelnuovo-Severi inequality implies that u is the unique map of degree e to $\mathbb{P}^1$ up to automorphisms of $\mathbb{P}^1$, so $u \circ s = \beta \circ u$ for some $\beta \in \mathrm{PGL}_2(\mathbb{Q})$ with $\beta^2 = 1$. If $\beta = 1$, then u factors through $W \to X_0(61)$, which is impossible as $e \leq 3$. Otherwise u induces a map $X_0(61) \to \mathbb{P}^1/\langle\beta\rangle \simeq \mathbb{P}^1$ of degree e, which is again impossible. □

6.4. **The prime 73.**

Proposition 6.8. *We have $73 \notin S(d)$ for $9 \leq d \leq 22$ if Z is a sum of points SS_6, NC_9 and SS_{18}.*

Proof. Let $h = \langle 2\rangle$, of order 9. It fixes each of the 4 supersingular closed points of degree 6 and the 4 non-rational cusps (of degree 9), and the points of degree 18, and the subgroup $\langle 8\rangle$ of order 3 fixes the geometric points of degree 6. We compute that h acts trivially on the common kernel of A_3 and A_5 on $H_1(X_1(73), \mathbb{Z}) \otimes \mathbb{F}_2$, which contains $J_1(73)(\mathbb{Q})[2]$, so $y = 0$ in Lemma 2.7 ii), and $\operatorname{gon}_{\mathbb{Q}} X_1(73) > 26$. As $d < 24 = (73-1)/3$, the orbits of $\langle 2\rangle$ on the support of D have length 9 (Lemma 2.2). Let $e = \deg Z$, so $c = d - e$ and $e \in \{6, 9, 12, 15, 18, 21\}$.

Suppose first that $d + c = 2d - e \leq 26$. Then Lemma 2.7 gives $hD = D$ and $9 \mid c$, and as the orbits of h have length 9, 9 divides d. Hence $d \in \{9, 18\}$ and D is a pullback from $Y = X_1(73)/\langle 2\rangle$, of a divisor D' of degree $d/9$. If $d = 9$, then the image of D' on $X_0(73)$ is a non-cuspidal rational point, which does not exist [23]. If $d = 18$, then D' maps to a non-cuspidal rational or quadratic point on $X_0(73)$. These were determined by Box [7, Section 4.7], and they have 11 different j-invariants (3 of them quadratic and not CM, and 8 rational CM j-invariants). For each of them we check, using the action of Frobenius on $E[73]$ at a prime which splits in the corresponding quadratic field, that no point of Y of degree at most 2 lies above it.

Suppose now that $2d - e > 26$. Then $d \geq 17$, and Z and d are as in Table 8. In these cases we use Lemma 2.10 with $t = ((h-1)A_3, (h-1)A_5)$ and

$$L = Z_0 + R + \pi^* Y^{(r)},$$

where Z_0 is the divisor of degree $6k$ whose points are the unique lifts of the geometric points of SS_6 in Z that are fixed by $\langle 8\rangle$ (here k is the number of points SS_6 in Z), $C' = \overline{R} + \pi^* E'$ with E' a sum of r' cusps of Y, R has no full orbit of $\langle 2\rangle$ in its support, and $r = r'$ plus the number of points NC_9 in Z (which are pullbacks of cusps of Y). We compute (NC_9, two or three non-ordinary points) that the rank condition holds in all cases, so D lies on L. If $R \neq 0$, then D contains a cusp, which is impossible. If $R = 0$ and $k > 0$, then D contains a point

fixed by $\langle 8\rangle$, which contradicts Lemma 2.1. If $R = 0$ and $k = 0$, then $Z = NC_9$, $d = 18$, and D is a pullback from Y, which we have already excluded. □

d	Z
17	SS_6
18	SS_6, NC_9
19	SS_6, NC_9
20	SS_6, NC_9, $2SS_6$
21	SS_6, NC_9, $2SS_6$, $SS_6 + NC_9$
22	SS_6, NC_9, $2SS_6$, $SS_6 + NC_9$

TABLE 8. The reductions at 73 for which we use Lemma 2.10.

6.5. **The primes** 89 **and** 97**.**

Proposition 6.9. *We have* $89 \notin S(d)$ *for* $11 \le d \le 22$ *if* $Z \in \{NC_{11}, 2NC_{11}, SS_{22}\}$.

Proof. Let $h = \langle 2\rangle$, of order 11, which fixes the 4 non-rational cusps. We compute that A_3 has odd degree on $(h-1)J_1(89)$, so $y = 0$ in Lemma 2.7 iii), and $\operatorname{gon}_{\mathbb{Q}} X_1(89) > 39$. If $Z = NC_{11}$, then $c = d - 11$ and $d + c \le 33$, so Lemma 2.7 gives $hD = D$ and $11 \mid c$. If $c = 0$, then $d = 11$, D is a single orbit of $\langle 2\rangle$ by Lemma 2.2, and its image on $X_0(89)$ is a non-cuspidal rational point, which does not exist. So $c = 11$ and $d = 22$.

It remains to deal with $d = 22$ and $Z \in \{NC_{11}, 2NC_{11}, SS_{22}\}$. The supersingular closed points of degree 22 form 5 orbits under $\Delta(89)$, of lengths $4, 4, 4, 2, 1$. For the first three, $\alpha A_3 Z$ or $\alpha A_5 Z$ is not principal (Proposition 5.3), and the other two are fixed by h. Hence, by Lemma 2.7 and Lemma 2.2, in all the remaining cases D is a pullback from $Y = X_1(89)/\langle 2\rangle$, of a divisor D' of degree 2. The curve Y has genus 27, and we compute that $\#Y(\mathbb{F}_2) = 8$ and $\#Y(\mathbb{F}_4) = 38$, so $\operatorname{gon}_{\mathbb{Q}} Y \ge \operatorname{gon}_{\mathbb{F}_4} Y \ge 8$.

If Z contains a non-rational cusp, then $\overline{D'}$ is a sum of two cusps of $Y_{\mathbb{F}_2}$, which are defined over $\mathbb{F}_2$. We compute that the map $Y^{(2)} \to J_Y$ given by $t = (h'-1)(T_q - \langle q\rangle - q)A_q$, where h' generates $\Delta(89)/\langle 2\rangle$, is a formal immersion at all the 36 sums of two such cusps. As t kills $J_Y(\mathbb{Q})$ and all the classes of differences of cusps, t maps D' and the cuspidal divisor lifting $\overline{D'}$ (which is defined over an unramified extension of $\mathbb{Q}_2$) to 0, so they are equal by [9, Lemma 4.2], which is a contradiction.

If $Z = SS_{22}$, let ι be the element of order 2 of $\Delta(89)/\langle 2\rangle$, which is cyclic of order 4. It fixes $\overline{D'}$, $(\iota - 1)J_Y(\mathbb{Q})$ is finite, and so, as in Lemma 2.7, $2\iota D' \sim 2D'$. As $\operatorname{gon}_{\mathbb{Q}} Y > 4$, it follows that $\iota D' = D'$. A point of Y fixed by ι lies below a point of $X_1(89)$ fixed by a non-trivial element of $\Delta(89)$, which has degree at least 30 by Lemma 2.1, so it has degree at least 3. Hence ι has no fixed points in the support of D', so $D' = y + \iota(y)$ for a point y of degree at most 2, and the image of D' on $Y/\langle\iota\rangle$ is twice a rational point. Its image on $X_0(89)$ is a non-cuspidal rational point, which does not exist. □

Proposition 6.10. *We have* $97 \notin S(d)$ *for* $16 \le d \le 22$ *if* $Z = SS_{16}$.

Proof. The two supersingular closed points of degree 16 are fixed by the subgroup $H_4 \subseteq \Delta(97)$ of order 4, and their geometric points are fixed by the subgroup H_3 of order 3. As the strange characters modulo 97 have order 12, $(h-1)J_1(97)(\mathbb{Q})$ is finite for $h \in H_4$, and Proposition 2.5 gives $\operatorname{gon}_{\mathbb{Q}} X_1(97) \ge 49$.

Let h be a generator of H_4. If $16 \le d \le 19$, then $c = d - 16 \le 3$ and $2(d+c) \le 44 < 49$, so Lemma 2.7 gives $hD = D$ and $4 \mid c$, so $c = 0$ and $d = 16$. As $16 < (97-1)/3$, D is a pullback from $X_1(97)/H_4$ (Lemma 2.2). Let H_8 be the subgroup of order 8, $W = X_1(97)/H_8$, of genus 41, and let h' be the element of order 3 of $\Delta(97)/H_8$. The image D_W of D on W is a closed point of degree $m \in \{2, 4\}$, and its reduction is fixed by h'. The group $(h'-1)J_W(\mathbb{Q})$ is finite, as J_W corresponds to the characters of order dividing 6, and we compute that A_3 has odd degree on $(h'-1)J_W$. As in Lemma 2.7, it follows that $h'D_W \sim D_W$, and as $\operatorname{gon}_{\mathbb{Q}} W \ge 49/8$ (a map $W \to \mathbb{P}^1$ of degree k gives a map $X_1(97) \to \mathbb{P}^1$ of degree $8k$), $h'D_W = D_W$. As $3 \nmid m$, h' fixes each geometric point of D_W, so the points of D are fixed by non-trivial elements of $\Delta(97)$, which contradicts Lemma 2.1.

If $20 \le d \le 22$, we use Lemma 2.10 with $Y = X_1(97)/H_3$ and $L = Z_0 + R + \pi^* Y^{(r)}$, where Z_0 is the divisor of degree 16 whose points are the lifts of the geometric points of SS_{16} fixed by H_3, and R and r are as in the proof of Proposition 6.8. Here we use $t = (h_3 - 1)\beta_n A_q$, where h_3 generates H_3, q is odd, and $\beta_n = T_n - a_n$, where $a_n \in \mathbb{Z}[\Delta(97)]$ is such that β_n kills the abelian subvariety of $J_1(97)$ corresponding to the strange characters (on which T_n acts through $\mathbb{Z}[\Delta(97)]$, as this subvariety has dimension 1 over the field of values of the characters). Then t kills $J_1(97)(\mathbb{Q})$ and $\pi^* J_Y$, and we compute that the rank condition holds. So D contains R, which is impossible if $R \ne 0$, or Z_0, which contradicts Lemma 2.1. □

6.6. **Reductions supported on the rational cusps at** 101**,** 103 **and** 131**.**

Proposition 6.11. *Let* (d, p) *be* $(20, 101)$*,* $(21, 101)$*,* $(22, 101)$*, or* $p = 103$ *and* $17 \le d \le 22$*, or* $p = 131$ *and* $18 \le d \le 22$*. Then* $Z \ne 0$*.*

Proof. For $p = 101$ we use $t = (h-1)\beta_n A_q$ with h a generator of $\Delta(101)$ and β_n as in the proof of Proposition 6.10, for $n \in \{2, 3, 5, 7\}$, and we compute that f is a formal immersion at all sums of rational cusps of degree d that do not contain all the 50 rational cusps, which is all of them.

Let $p = 103$ (resp. $p = 131$), let $H \subseteq \Delta(p)$ be the subgroup of order 3 (resp. 5), and $Y = X_H$. Then $Y \to X_0(p)$ has degree 17 (resp. 13) and the rational cusps of Y form one fibre F of this map. On Y, the criterion of Proposition 5.1 holds at all sums of rational cusps of degree d that do not contain F. If $\overline{\pi_* D}$ contains F, write $\overline{\pi_* D} = F + \overline{R}$. We use Lemma 2.10 on $Y^{(d)}$ with $L = R + \rho^* X_0(p)$, where $\rho : Y \to X_0(p)$, and $t = (\langle a \rangle - 1)(T_3 - \langle 3 \rangle - 3)$ on J_Y, where $\langle a \rangle$ generates $\Delta(p)/H$, which kills $J_Y(\mathbb{Q})$ (as p is not strange) and $\rho^* J_0(p)$. We compute that the rank condition holds, so $\pi_* D$ lies on L. If $R \ne 0$, then $\pi_* D$ contains a cusp, which is impossible. If $R = 0$, then $p = 103$, $d = 17$, and $\pi_* D$ is the pullback of a non-cuspidal rational point on $X_0(103)$, which does not exist. □

6.7. **The primes** 109**,** 113**,** 127 **and** 137**.** At these primes we only use $2y = 0$ in Lemma 2.7, so we need $\operatorname{gon}_{\mathbb{Q}} X_1(p) > 2(d+c)$, and $d < (p-1)/3$, so D is a pullback from X_H whenever it is H-invariant.

Proposition 6.12. *We have* $109 \notin S(d)$ *for* $12 \le d \le 22$ *if* $Z \in \{SS_9, NC_{18}, 2SS_9, SS_9 + SS_{12}\}$*.*

Proof. All of these are fixed by $h = \langle 23 \rangle$, of order 18, and $\operatorname{gon}_{\mathbb{Q}} X_1(109) > 58$. If $Z = SS_9$ and $d \le 19$, or if Z is one of the others, then $2(d+c) \le 58$, so $hD = D$ and $18 \mid c$. As $c \le 13$, this gives $c = 0$, so $Z = NC_{18}$ or $2SS_9$ with $d = 18$, or $Z = SS_9 + SS_{12}$ with $d = 21$. In the last case $18 \nmid 21$, which contradicts Lemma 2.2, and in the first two D is a single orbit of $\langle 23 \rangle$, whose image on $X_0(109)$ is a non-cuspidal rational point. If $Z = SS_9$ and $d \ge 20$, we use Lemma 2.10 with $L = C + \pi^* Y$ or $L = R + \pi^* Y^{(2)}$, where $Y = X_1(109)/\langle h^2 \rangle$ and $\langle h^2 \rangle$, of

order 9, acts freely on the geometric points of SS_9, and $t = (h^2 - 1)A_q$ (code). As before, D contains a cusp, which is a contradiction. □

Proposition 6.13. *We have* $113 \notin S(d)$ *for* $13 \leq d \leq 22$ *if* Z *is one of the divisors listed for* 113 *in Table 7.*

Proof. Here O_8 denotes one of the 7 ordinary closed points of degree 8 which are fixed by $h_8 = \langle 40 \rangle$, of order 8, and we have $\mathrm{gon}_{\mathbb{Q}} X_1(113) > 63$. The element $h_2 = \langle 2 \rangle$, of order 14, fixes the closed points SS_7 and NC_{14}, and the involution $\iota = \langle 98 \rangle$ fixes the points SS_7 and O_8.

If $Z \in \{2SS_7, 3SS_7, NC_{14}, SS_7 + NC_{14}\}$, then $2(d + c) \leq 60$, so $h_2 D = D$ and $14 \mid c$, so $c = 0$. Then $d \in \{14, 21\}$, and as $14 \nmid 21$, $d = 14$ and D is a single orbit of h_2, whose image on $X_0(113)$ is a non-cuspidal rational point, which is impossible.

If $Z = O_8$ and $d \leq 19$, or $Z = 2O_8$, then $2(d+c) \leq 62$, so $h_8 D = D$ and $8 \mid c$. Hence $d = 16$ and $c = 0$ or $c = 8$, and D is a pullback of a divisor of degree 2 from $X_1(113)/\langle h_8 \rangle$. Its image on $X_0(113)$ is a non-cuspidal point of degree at most 2, which has complex multiplication, as all the quadratic points on $X_0(113)$ are CM points [3, Table 14] and there are no non-cuspidal rational points. But then by Silverberg's theorem (see the proof of Lemma 2.1), a point of order 113 on E is defined over a field of degree at least $112/6 > 16$, which is a contradiction. If $Z = O_8$ and $d \geq 20$, we use Lemma 2.10 with $L = R + \pi^* Y^{(r)}$, where π is the quotient by $\langle h_8 \rangle$ near Z, which is étale there, and $t = ((h_8 - 1)A_2, (h_8 - 1)A_3)$ (code).

If $Z = O_8 + SS_7$, then $2(d+c) \leq 58$, so $\iota D = D$ and $2 \mid c = d - 15$. So d is odd, and the orbits of ι on D have length 2 by Lemma 2.2, which is a contradiction. Finally, if $Z = O_8 + NC_{14}$ and $d = 22$, we use Lemma 2.10 with $t = (h_8 - 1)A_q A_{q'}$, and $L = Z_1 + \pi^* Y$, where Z_1 is the lift of NC_{14} to a cuspidal divisor (code). Hence D contains a cusp, which is a contradiction. □

Proposition 6.14. *We have* $127 \notin S(d)$ *for* $17 \leq d \leq 22$ *if* Z *is one of the divisors listed for* 127 *in Table 7.*

Proof. The element $h_7 = \langle 2 \rangle$, of order 7, fixes the 9 non-rational cusps and the 48 supersingular closed points of degree 14, and the ordinary closed points O_9 are fixed by the subgroup H_9 of order 9. We have $\mathrm{gon}_{\mathbb{Q}} X_1(127) > 79$, and A_2 kills $J_1(127)(\mathbb{Q})_{\mathrm{tors}}$ by Lemma 2.4 b).

Suppose that Z is a sum of points NC_7 and SS_{14}. Then $2(d + c) \leq 74$, so $h_7 D = D$ and $7 \mid c$. This is impossible unless $d = 21$, in which case D is a pullback from $Y = X_1(127)/\langle 2 \rangle$, of a divisor D' of degree 3. The curve Y has genus 88, and its points over $\mathbb{F}_2$ which are images of cusps are the 9 rational cusps and the 9 images of the non-rational cusps. If Z consists of non-rational cusps, then $\overline{D'}$ is a sum of three of these 18 points, and we compute that the map $Y^{(3)} \to J_Y$ given by $t = (h' - 1)(T_q - \langle q \rangle - q)A_q$, $q \in \{2, 3\}$, is a formal immersion at all the 1140 such sums, which shows, as in the proof of Proposition 6.9, that D' is cuspidal. If $Z = SS_{14} + C_7$, then for 5 of the 6 orbits of points SS_{14}, $\alpha A_q Z$ is not principal for some $q \in \{3, 5, 7\}$, and for the remaining orbit (of length 3), $\overline{D'} = S + \overline{c_1}$, where c_1 is a rational cusp of Y and S is a supersingular point of degree 2 fixed by the element h_3' of order 3 of $\Delta(127)/\langle 2 \rangle$. We use Lemma 2.10 on $Y^{(3)}$ with $L = S_0 + c_1$, where S_0 is the unique lift of S fixed by h_3', and $t = (h_3' - 1)(T_q - \langle q \rangle - q)$ (code), so D' contains a cusp. The case $SS_{14} + NC_7$ is excluded by Proposition 5.3.

If $Z = O_9$ or $Z = 2O_9$, then $2(d + c) \leq 72$, so D is H_9-invariant and $9 \mid c$. Hence $d = 18$ and D is a pullback of a divisor of degree 2 from $X_1(127)/H_9$. Its image on $X_0(127)$ is a non-cuspidal point of degree at most 2, which is a CM point by [3, Table 16], and as in the proof of Proposition 6.13 this contradicts Silverberg's theorem, as $126/6 = 21 > 18$.

Finally, if $Z = O_9 + NC_7$, we use Lemma 2.10 with $L = C + \pi_9^*(X_1(127)/H_9) + \pi_7^*(X_1(127)/\langle 2 \rangle)$, which is smooth of dimension 2 as H_9 and $\langle 2 \rangle$ act freely on the supports of

O_9 and NC_7, and $t = (h_9 - 1)(h_7 - 1)A_q$, $q \in \{2, 3\}$, where h_9 generates H_9 (code). So D contains $C \neq 0$, which is a contradiction. □

Proposition 6.15. *We have* $137 \notin S(d)$ *for* $19 \leq d \leq 22$ *if* $Z \in \{SS_{17}, O_{17}\}$.

Proof. The point SS_{17} is fixed by $\langle 2 \rangle$, of order 34, and O_{17} is fixed by $\langle 81 \rangle$, of order 17. In both cases $c = d - 17 \in \{2, 3, 4, 5\}$ and $2(d + c) \leq 54 < \mathrm{gon}_{\mathbb{Q}} X_1(137)$, so Lemma 2.7 gives $34 \mid c$ or $17 \mid c$, which is a contradiction. □

6.8. **Proof of Theorem 1.1.** By Proposition 3.3 all the primes listed in Theorem 1.1 are in $S(d)$. Conversely, let $9 \leq d \leq 22$ and let $p \in S(d)$ be a prime which is not listed. By Propositions 4.12 and 4.13, p is one of the primes of Table 4. By Propositions 5.1, 5.3 and 6.11, the reduction of the trace divisor is one of those listed in Table 7, and these are excluded by Propositions 6.1, 6.2, 6.4 to 6.6, 6.8 to 6.10 and 6.12 to 6.15.

7. The computations

Most of the computations were done in `Magma` [6], version V2.30-1, using the package `mdmagma` of Derickx [25] and the models of $X_1(N)$ of Sutherland [36], and the independent checks were done in SageMath [37]. For the computations with modular symbols of large level in Lemma 4.11 we exported the sparse matrices of T_2 from `Magma` and computed the sequences in our own exact `C++` code, checking the first 64 terms against `Magma` in every case, and the certificates of linear complexity were checked independently with FLINT. All the computations were performed on the *Mordell* server at the Department of Mathematics, University of Zagreb, with 32 cores and 376 GB of RAM.

Data availability and reproducibility. All the code for the computations in this paper, the logs, the machine-readable ledgers and the certificates that fit in a repository are available at

`https://github.com/F-Najman/Sd`

All the links to the repository point to its revision `1caaabb`. In the digital version of this paper, the words reporting a computation, such as "we compute", are links to the corresponding script. The repository is organized by degree: the folder `audit11` contains the computations for $d \leq 11$ (and `audit10` the transfer to $d = 10$), and the folders `degree12`, ..., `degree22` those for the larger degrees. Each computation for degree d reuses the certificates for the smaller degrees, after checking that the closed points used there are the same, so the certificates for a given prime are spread over the folders of the degrees in which they were first needed. For each degree, the file `final_coverage.json` records for every prime up to the bound of Table 2 the reason it is excluded or the example showing that it occurs, `support_coverage.json` records the argument used for every possible type of reduction at every prime of Table 4, and the scripts `verify_coverage.py` check these ledgers against the certificates. The folder `proofs` contains the detailed accounts of the computations for each degree, with the numbers of divisors checked in each case, and the `README.md` file explains how to rerun everything. The complete certificates (about 360 GB, mostly the matrices of the principal divisor tests and the modular symbol computations) are too large for the repository. The repository contains their SHA256 hashes, and they are available from the author on request.

The computations were checked in the following ways. The ranks of all the matrices certifying that divisors are not principal (Lemma 2.8), that maps are formal immersions (Lemmas 2.9 and 2.10) and the intersection matrices of Lemma 4.1 were recomputed independently in SageMath from the stored matrices, and the relation spaces used in the formal immersion computations were checked to span the full spaces of relations. The census of Section 5.1 was

checked against independent point counts, and the list of types of reductions was produced twice, independently. The certificates of linear complexity were checked independently of the code that produced them. The computations for $d \leq 11$, which are also the basis of the computations for the larger degrees, were redone from scratch after a first version, and then checked in an independent review (see `proofs/S10_S11_independent_audit.txt`). Finally, Theorem 1.1 for $d = 9$ was also proved, during the work on [28], by an independent set of computations, which partly use quotients of $X_1(p)$ instead of $X_1(p)$. These are in `degree09`.

We should mention that in the first version of the computations for $d \in \{15, 16\}$ the element α at $p = 97$ was the polynomial $(\sigma^2 - 1)(\sigma^4 - 1)$ in a generator σ of $\Delta(97)$, which does not kill the strange characters of order 12, so Lemma 5.2 does not apply. We replaced it by $\sigma^{12} - 1$ and $(\sigma - 1)\Phi_{12}(\sigma)$ and redid all the affected computations (see `proofs/S97_correction.txt`). The computations for the other degrees were not affected.

References

[1] A. Abbes and E. Ullmo, *À propos de la conjecture de Manin pour les courbes elliptiques modulaires*, Compositio Math., 103 (1996), pp. 269–286.

[2] D. Abramovich, *A linear lower bound on the gonality of modular curves*, Internat. Math. Res. Notices, (1996), pp. 1005–1011.

[3] N. Adžaga, T. Keller, P. Michaud-Jacobs, F. Najman, E. Ozman, and B. Vukorepa, *Computing quadratic points on modular curves $X_0(N)$*, Math. Comp., 93 (2024), pp. 1371–1397.

[4] A. Agashe, K. Ribet, and W. A. Stein, *The Manin constant*, Pure Appl. Math. Q., 2 (2006), pp. 617–636.

[5] A. R. Booker, M. Lee, and A. Strömbergsson, *Twist-minimal trace formulas and the Selberg eigenvalue conjecture*, J. Lond. Math. Soc. (2), 102 (2020).

[6] W. Bosma, J. Cannon, and C. Playoust, *The Magma algebra system. I. The user language*, J. Symbolic Comput., 24 (1997), pp. 235–265. Computational algebra and number theory (London, 1993).

[7] J. Box, *Quadratic points on modular curves with infinite Mordell-Weil group*, Math. Comp., 90 (2021), pp. 321–343.

[8] M. Derickx, A. Etropolski, M. van Hoeij, J. S. Morrow, and D. Zureick-Brown, *Sporadic cubic torsion*, Algebra Number Theory, 15 (2021), pp. 1837–1864.

[9] M. Derickx, S. Kamienny, W. Stein, and M. Stoll, *Torsion points on elliptic curves over number fields of small degree*, Algebra Number Theory, 17 (2023), pp. 267–308.

[10] M. Derickx and F. Najman, *Classification of torsion of elliptic curves over quartic fields*, J. Reine Angew. Math., 829 (2025), pp. 123–156.

[11] M. Derickx and M. Stoll, *Prime order torsion on elliptic curves over number fields. Part I: Asymptotics.* Preprint, https://arxiv.org/abs/2505.14109v1, 2025.

[12] ———, *Prime order torsion on elliptic curves over number fields. Part II.* Unpublished preprint, 2026.

[13] M. Derickx and M. van Hoeij, *Gonality of the modular curve $X_1(N)$*, J. Algebra, 417 (2014), pp. 52–71.

[14] Y. Hasegawa and M. Shimura, *Trigonal modular curves*, Acta Arith., 88 (1999), pp. 129–140.

[15] D. Jeon, *Modular curves with infinitely many cubic points*, J. Number Theory, 219 (2021), pp. 344–355.

[16] S. Kamienny, *Torsion points on elliptic curves and q-coefficients of modular forms*, Invent. Math., 109 (1992), pp. 221–229.

[17] S. Kamienny and B. Mazur, *Rational torsion of prime order in elliptic curves over number fields*, Astérisque, (1995), pp. 81–100. With an appendix by A. Granville.

[18] K. Kato, *p-adic Hodge theory and values of zeta functions of modular forms*, Astérisque, (2004), pp. ix, 117–290. Cohomologies p-adiques et applications arithmétiques. III.

[19] N. M. Katz, *Galois properties of torsion points on abelian varieties*, Invent. Math., 62 (1981), pp. 481–502.

[20] M. A. Kenku and F. Momose, *Torsion points on elliptic curves defined over quadratic fields*, Nagoya Math. J., 109 (1988), pp. 125–149.

[21] M. Khawaja, *Torsion primes for elliptic curves over degree 8 number fields*, Res. Number Theory, 10 (2024), p. Paper No. 48.

[22] B. Mazur, *Modular curves and the Eisenstein ideal*, Inst. Hautes Études Sci. Publ. Math., (1977), pp. 33–186 (1978). With an appendix by Mazur and M. Rapoport.

[23] B. Mazur, *Rational isogenies of prime degree*, Invent. Math., 44 (1978), pp. 129–162.

[24] B. Mazur and P. Swinnerton-Dyer, *Arithmetic of Weil curves*, Invent. Math., 25 (1974), pp. 1–61.
[25] *mdmagma*. https://github.com/koffie/mdmagma. M. Derickx, `Magma` package.
[26] L. Merel, *Bornes pour la torsion des courbes elliptiques sur les corps de nombres*, Invent. Math., 124 (1996), pp. 437–449.
[27] F. Najman, *Classification of torsion of elliptic curves over quintic fields*. Preprint, https://arxiv.org/abs/2609.12846, 2026.
[28] ———, *Torsion of elliptic curves over number fields of degree* 6*,* 7*,* 8 *and* 9. Preprint, 2026.
[29] F. Najman and M. Varivoda, *Torsion groups of elliptic curves that appear infinitely often over septic, octic and nonic fields*. Preprint, https://arxiv.org/abs/2602.03513, 2026.
[30] P. Parent, *Bornes effectives pour la torsion des courbes elliptiques sur les corps de nombres*, J. Reine Angew. Math., 506 (1999), pp. 85–116.
[31] ———, *Torsion des courbes elliptiques sur les corps cubiques*, Ann. Inst. Fourier (Grenoble), 50 (2000), pp. 723–749.
[32] ———, *No* 17*-torsion on elliptic curves over cubic number fields*, J. Théor. Nombres Bordeaux, 15 (2003), pp. 831–838.
[33] D. Prasad and C. S. Yogananda, *Bounding the torsion in CM elliptic curves*, C. R. Math. Acad. Sci. Soc. R. Can., 23 (2001), pp. 1–5. Preprint version https://arxiv.org/abs/math/9810208.
[34] S. Siksek, *Chabauty for symmetric powers of curves*, Algebra Number Theory, 3 (2009), pp. 209–236.
[35] A. Silverberg, *Torsion points on abelian varieties of CM-type*, Compositio Math., 68 (1988), pp. 241–249.
[36] A. V. Sutherland, *Constructing elliptic curves over finite fields with prescribed torsion*, Math. Comp., 81 (2012), pp. 1131–1147.
[37] The Sage Developers, *SageMath, the Sage Mathematics Software System (Version 10.7)*. https://www.sagemath.org, 2025.
[38] M. van Hoeij, *Low degree places on the modular curve* $X_1(N)$. Preprint, https://arxiv.org/abs/1202.4355; data at https://www.math.fsu.edu/~hoeij/files/X1N/, 2014.
[39] M. Varivoda, *Gonality and quintic points on intermediate modular curves*. Preprint, https://hexagonmath.org/2610.00219v1, 2026.

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